

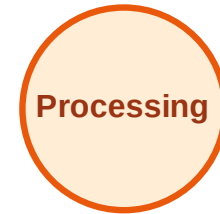
BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing

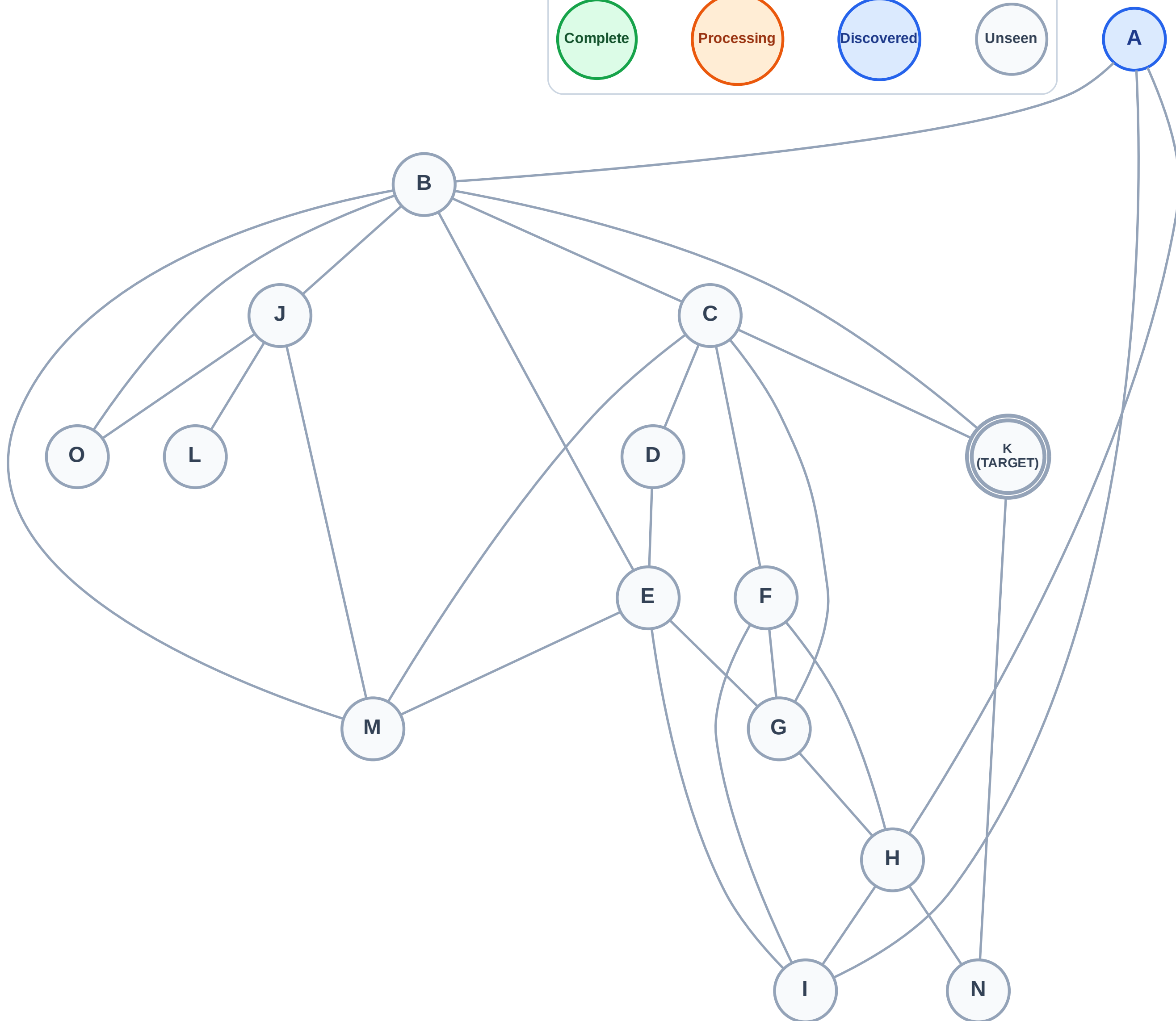


Discovered



Unseen

Queue: A
Visited: (none)



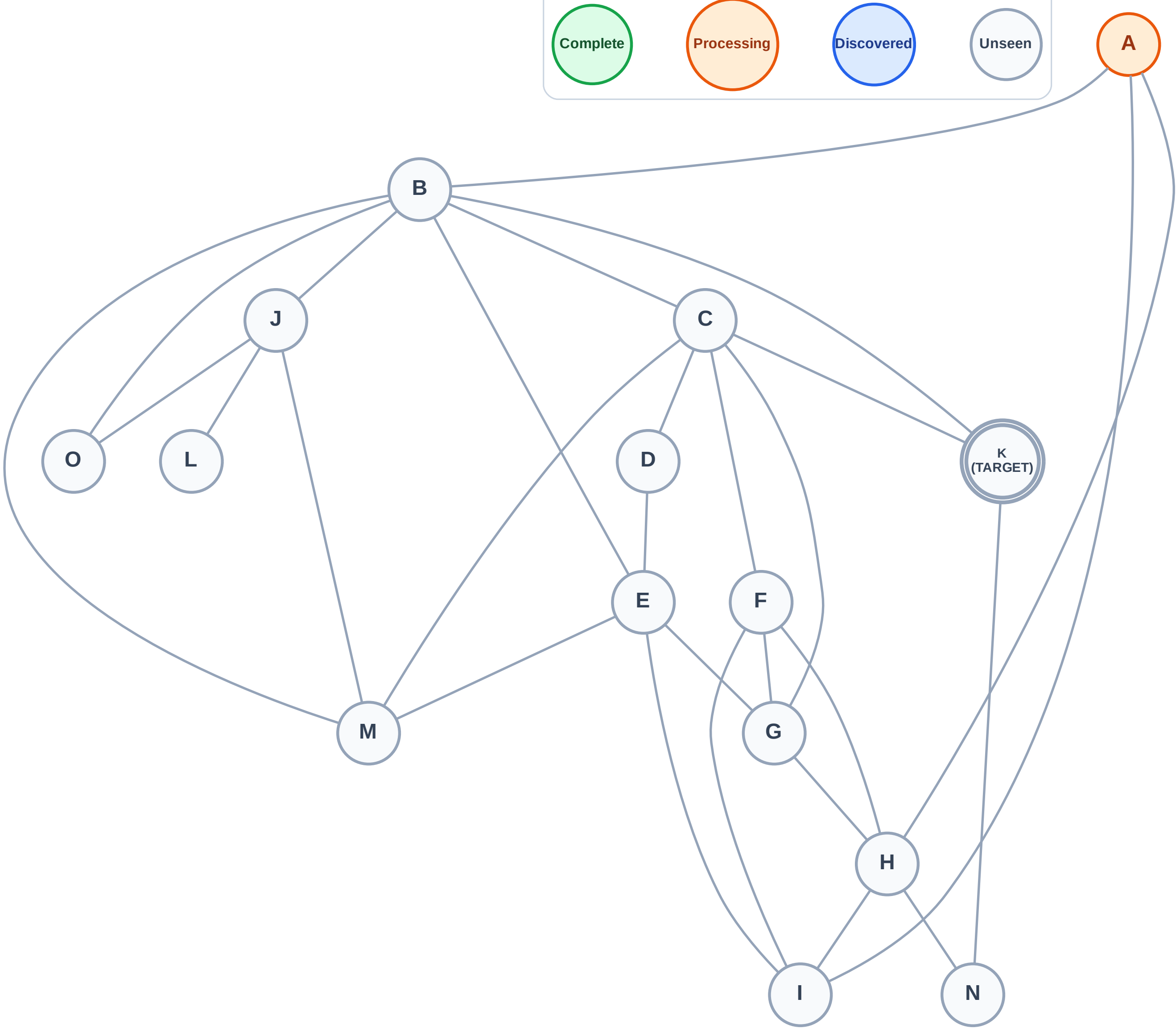
BFS Traversal

Step 2: Process node A

Legend



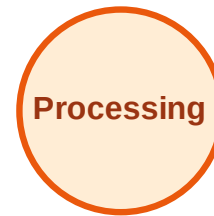
Queue: (empty)
Visited: (none)



BFS Traversal

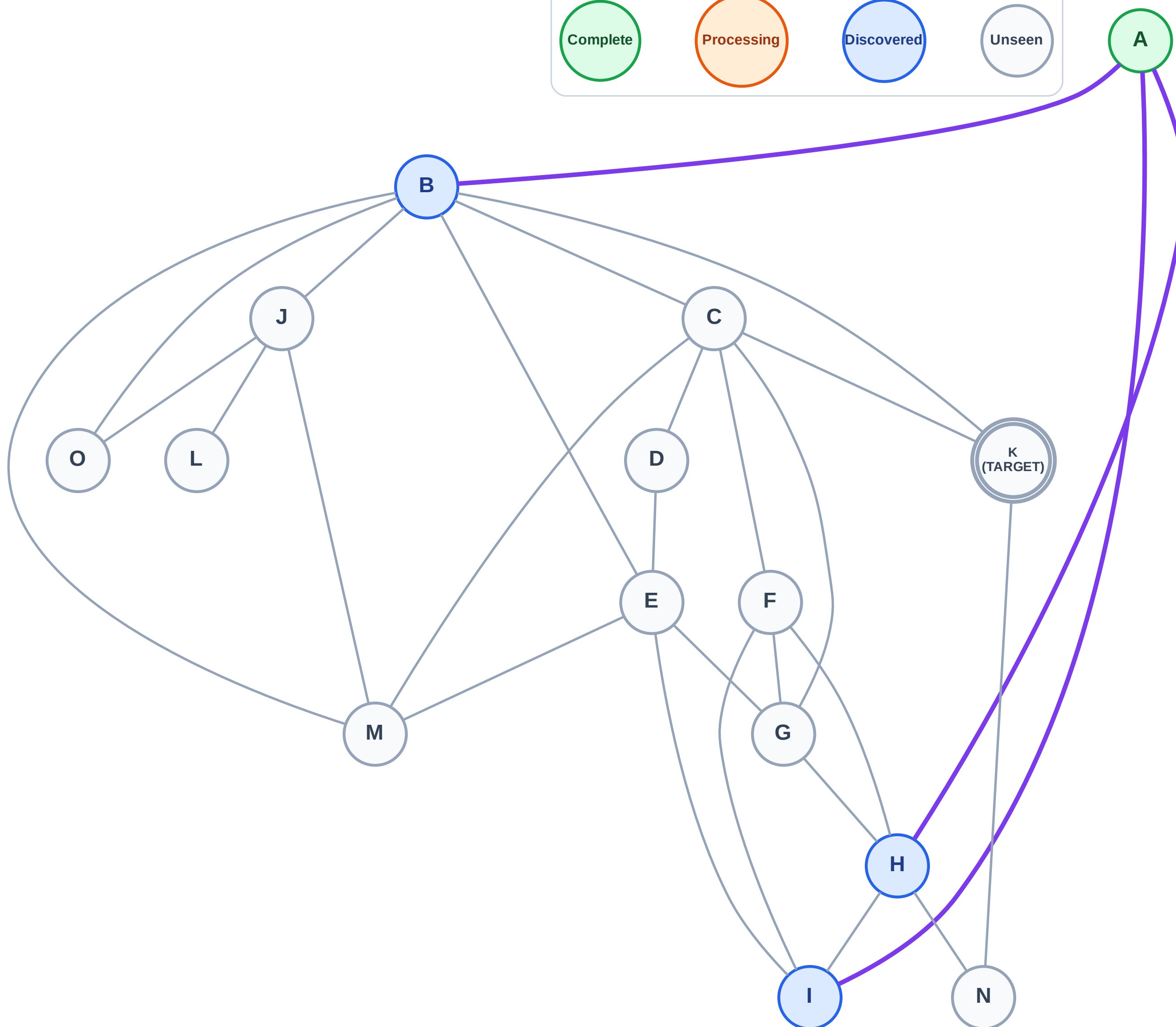
Step 3: Discover B, H, I from A

Legend



Queue: B → H → I

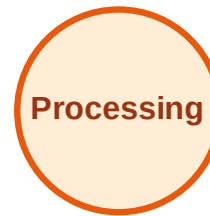
Visited: A



BFS Traversal

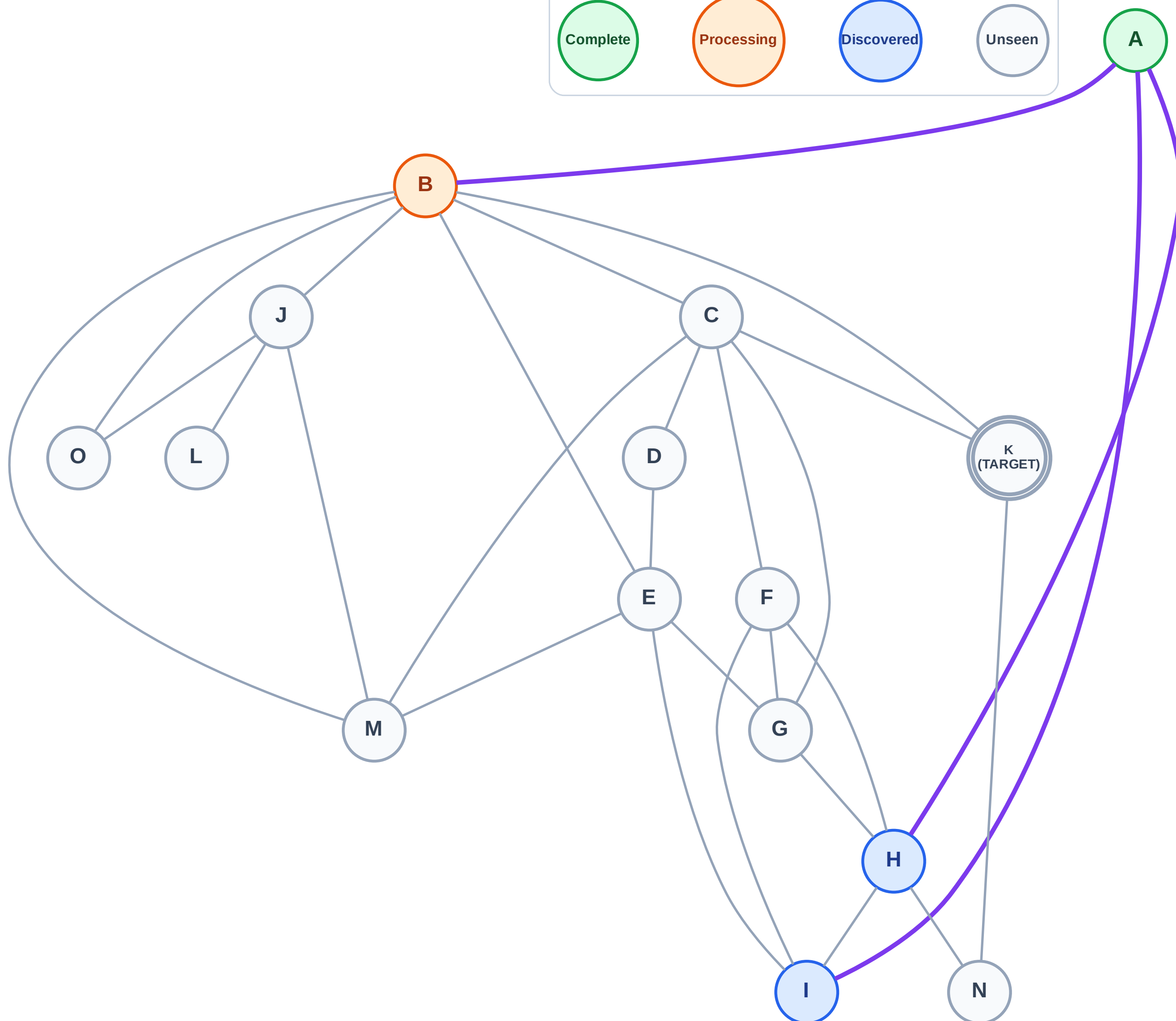
Step 4: Process node B

Legend



Queue: H → I

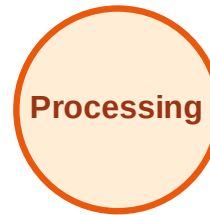
Visited: A



BFS Traversal

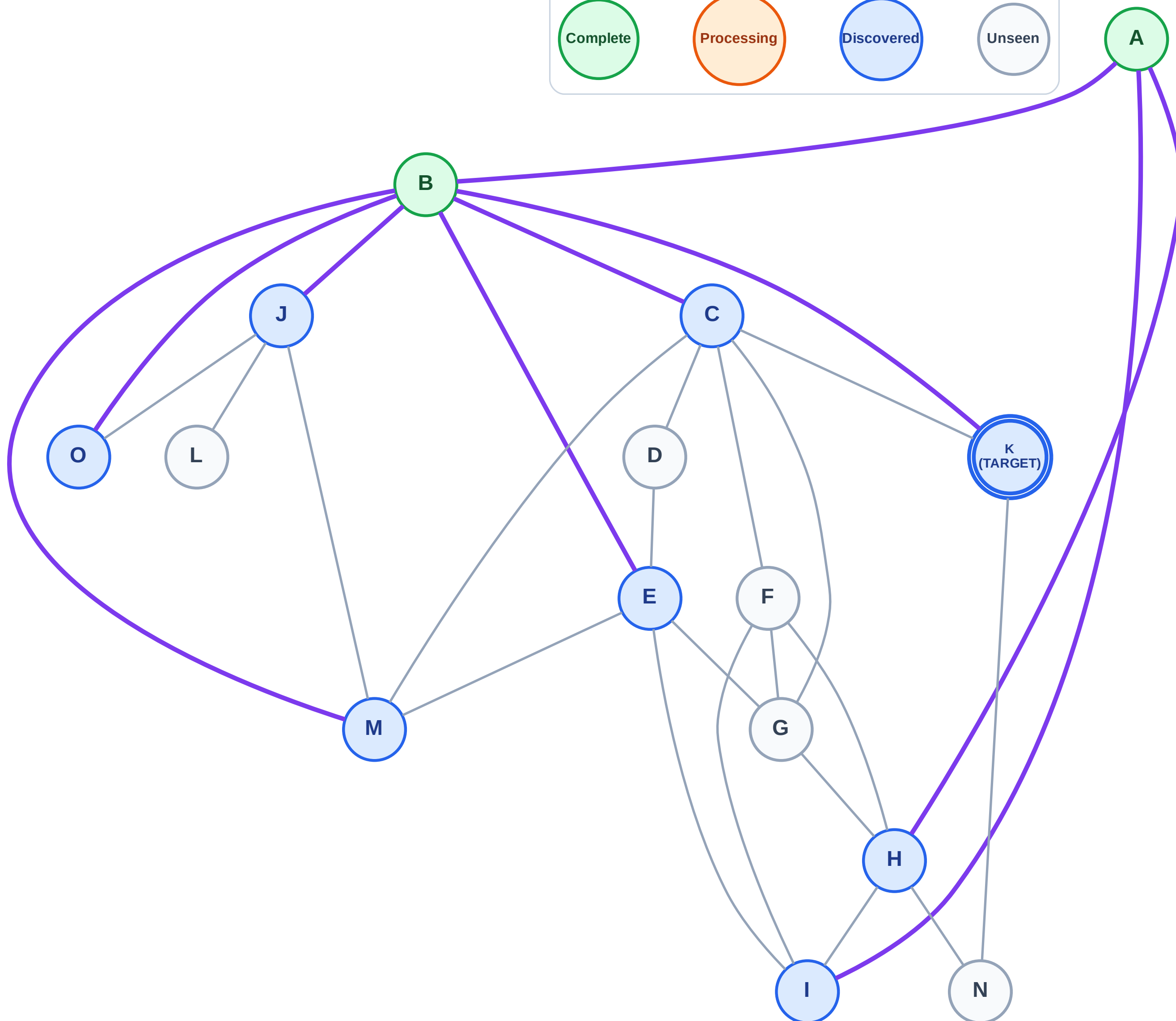
Step 5: Discover C, E, J, K, M, O from B

Legend



Queue: H → I → C → E → J → K → M → O

Visited: A, B



BFS Traversal

Step 6: Process node H

Legend

Complete

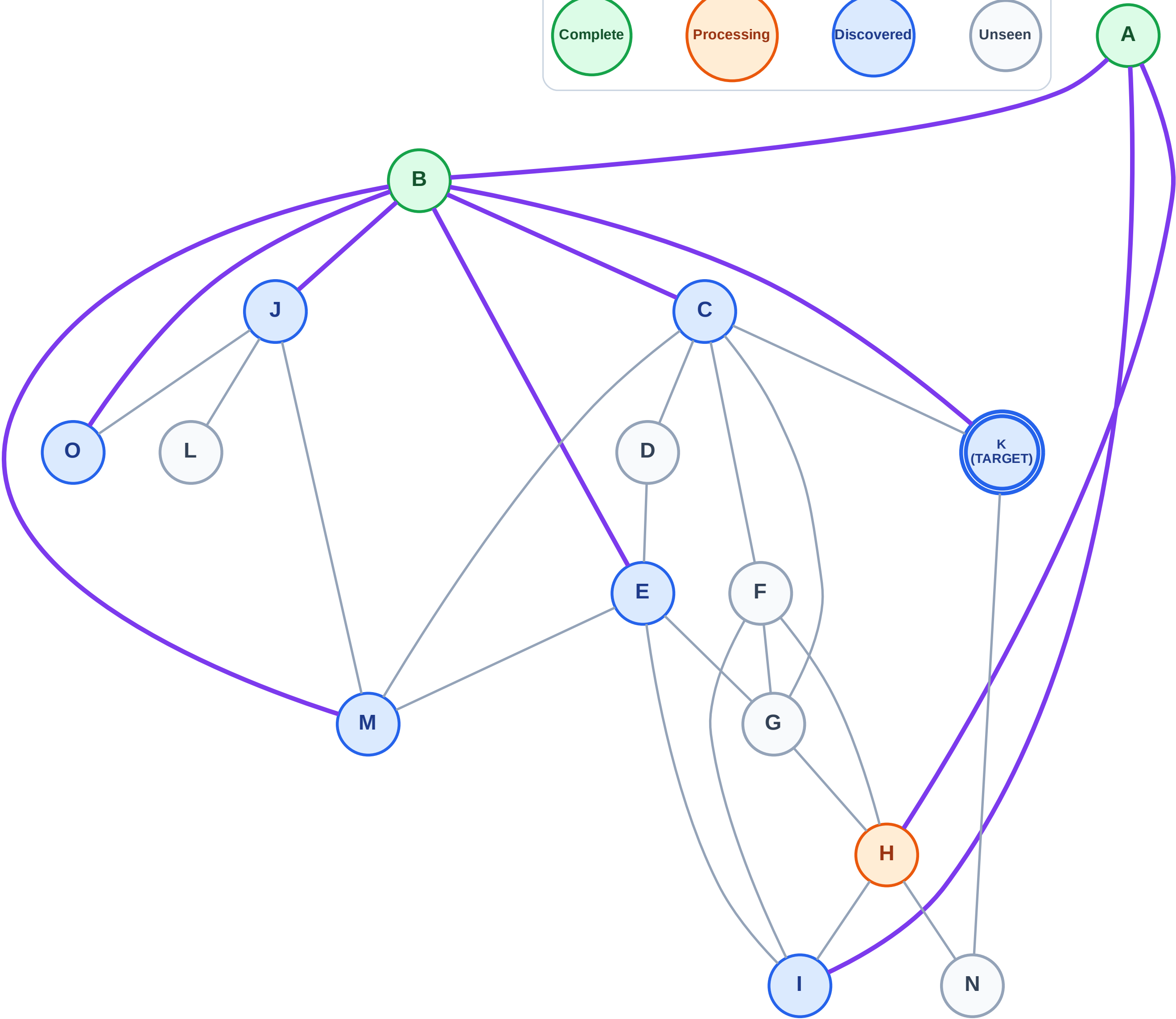
Processing

Discovered

Unseen

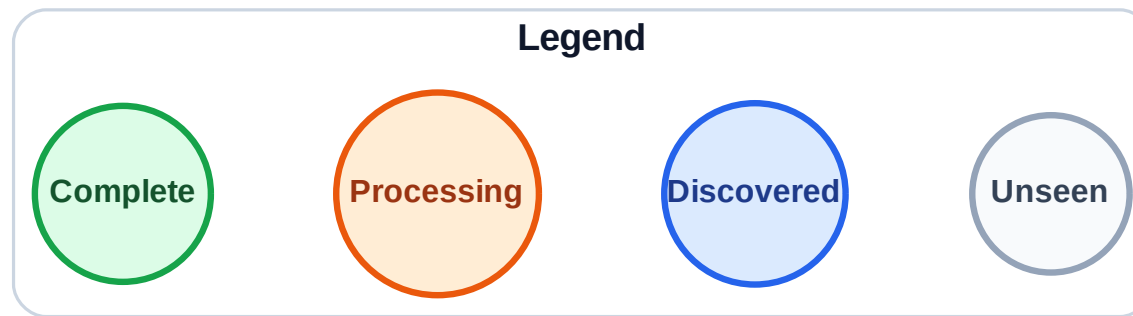
Queue: I → C → E → J → K → M → O

Visited: A, B

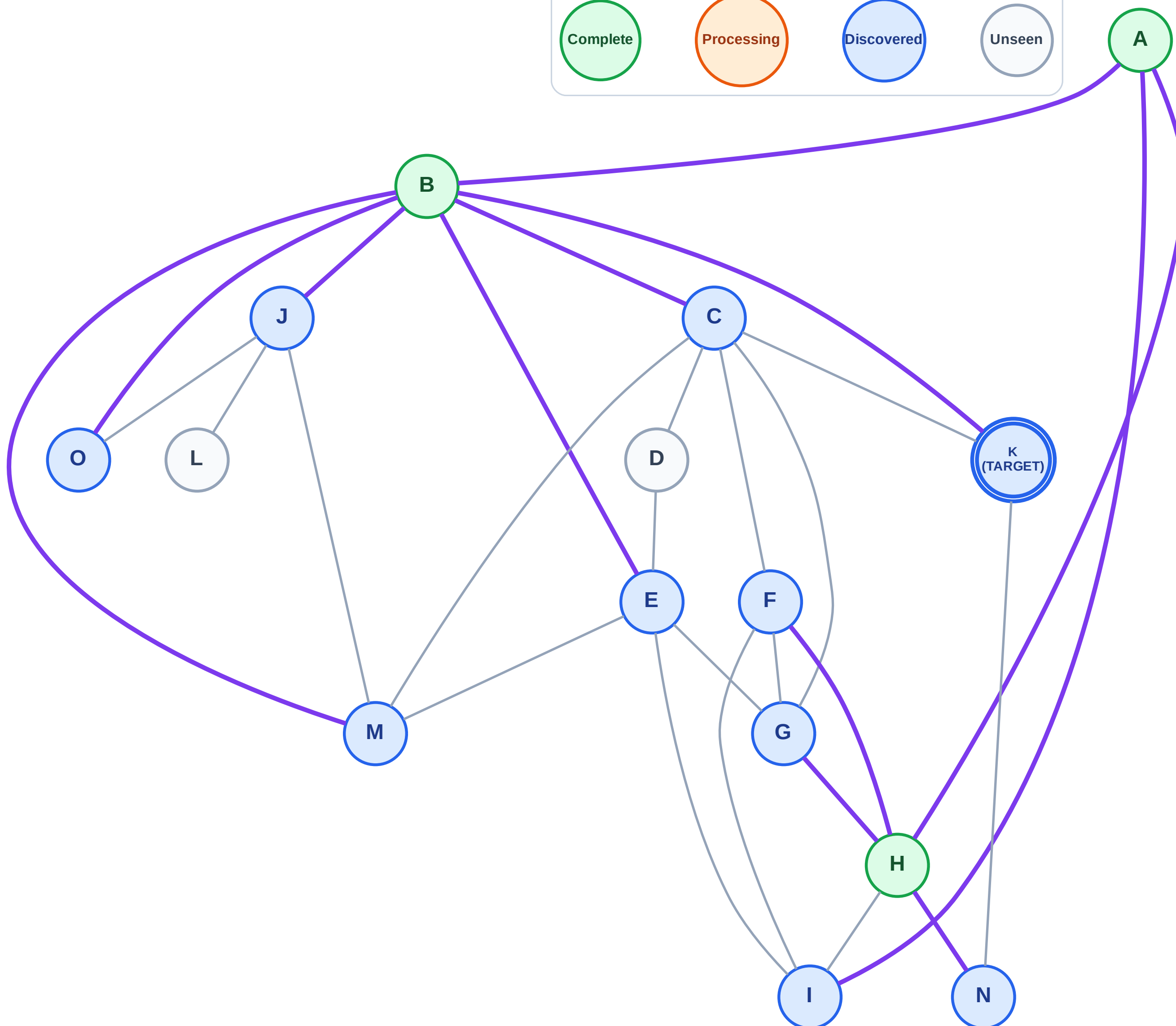


BFS Traversal

Step 7: Discover F, G, N from H



Queue: I → C → E → J → K → M → O → F → G → N
Visited: A, B, H



BFS Traversal

Step 8: Process node I

Legend

Complete

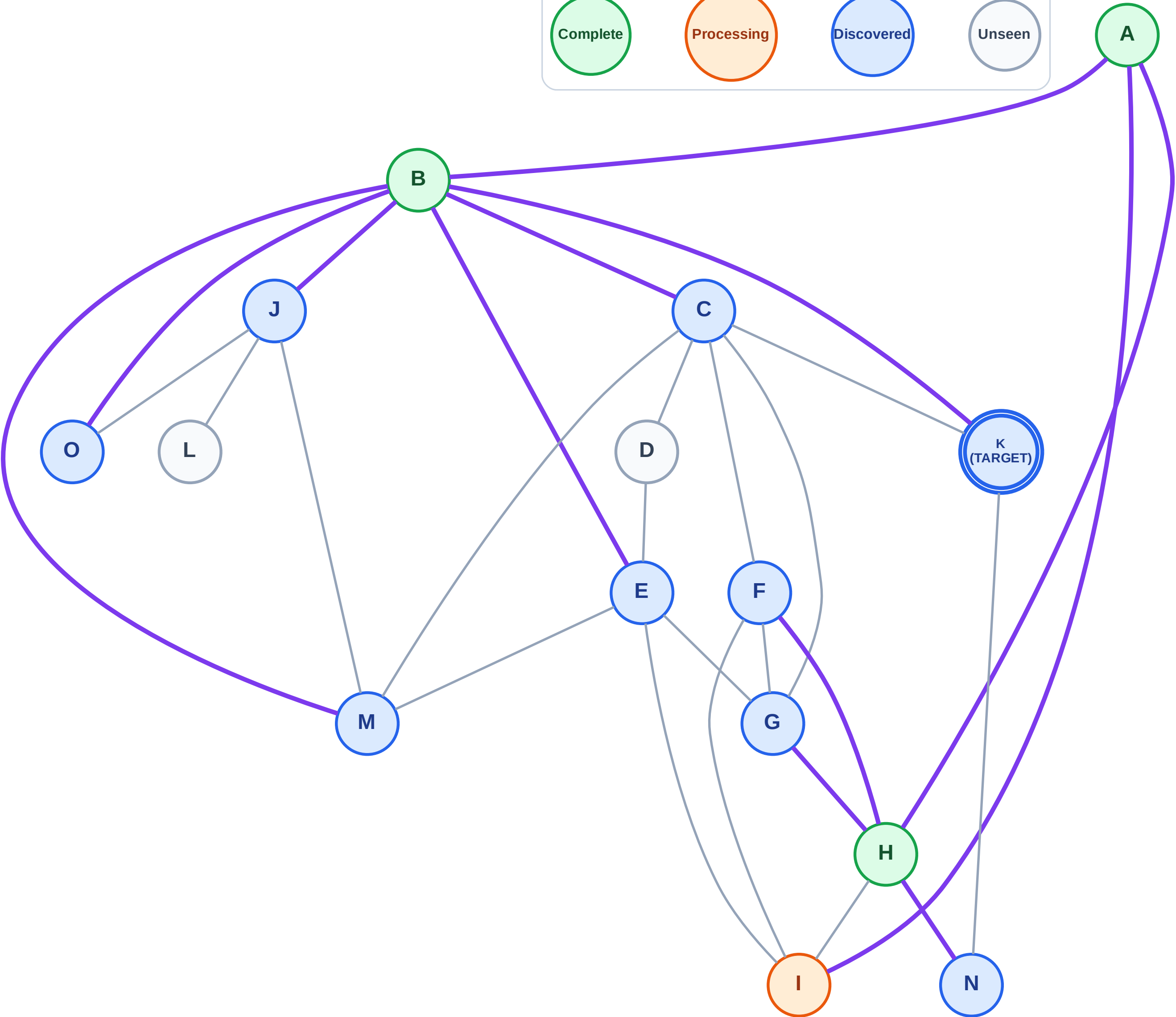
Processing

Discovered

Unseen

Queue: C → E → J → K → M → O → F → G → N

Visited: A, B, H



BFS Traversal

Step 9: No new neighbors from I

Legend

Complete

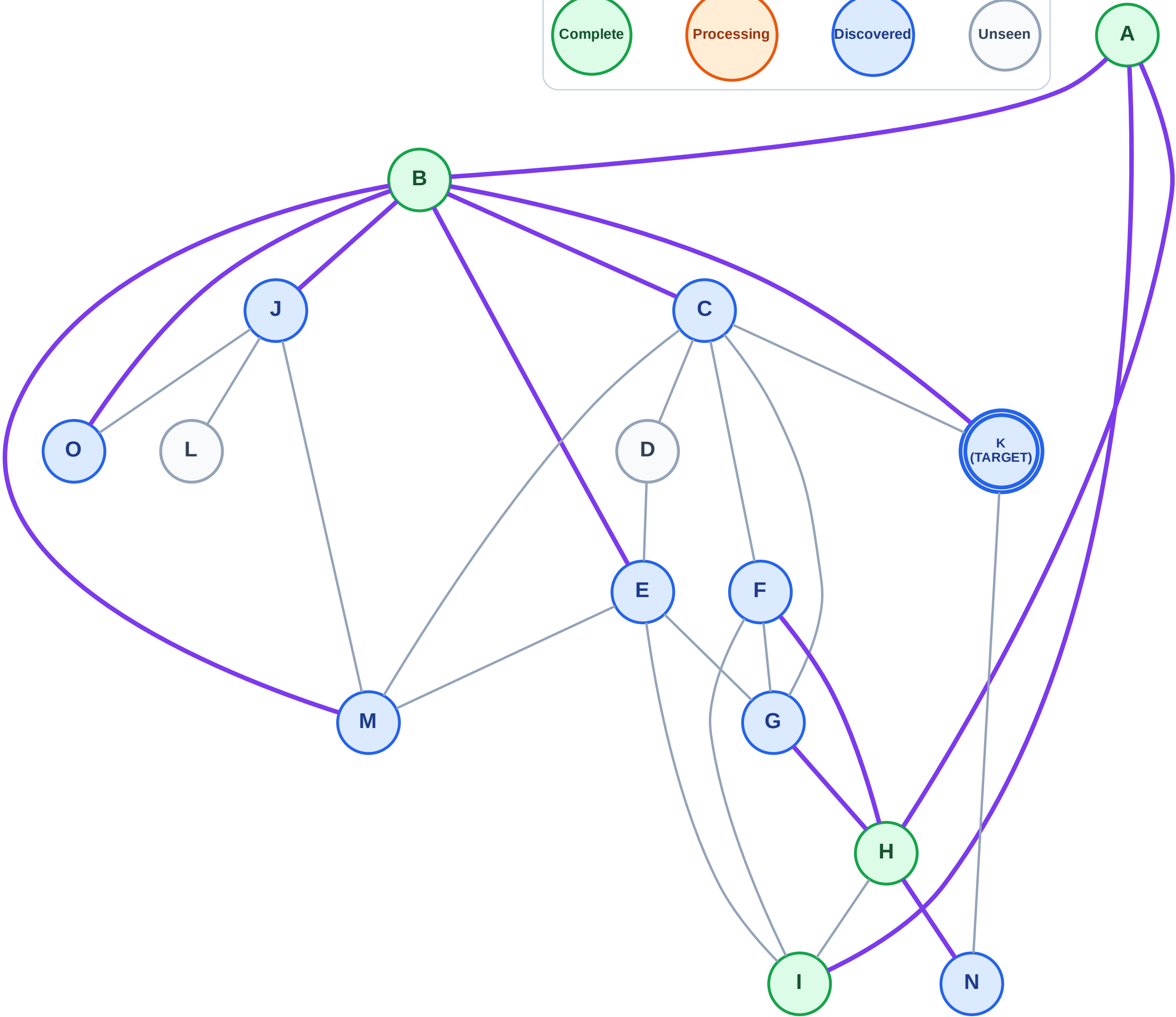
Processing

Discovered

Unseen

Queue: C → E → J → K → M → O → F → G → N

Visited: A, B, H, I



BFS Traversal

Step 10: Process node C

Legend

Complete

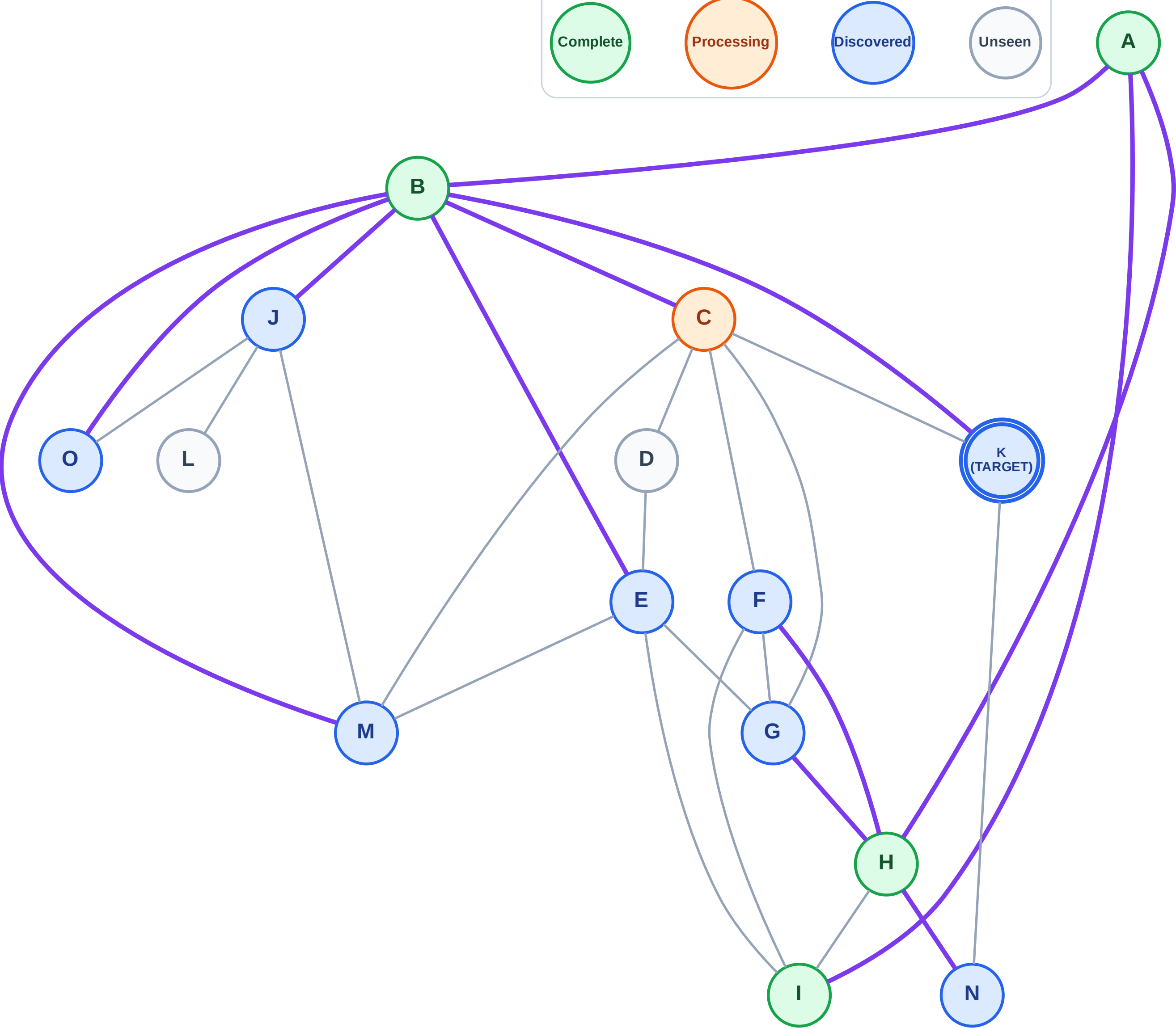
Processing

Discovered

Unseen

Queue: E → J → K → M → O → F → G → N

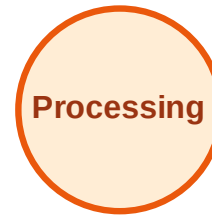
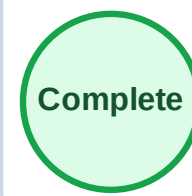
Visited: A, B, H, I



BFS Traversal

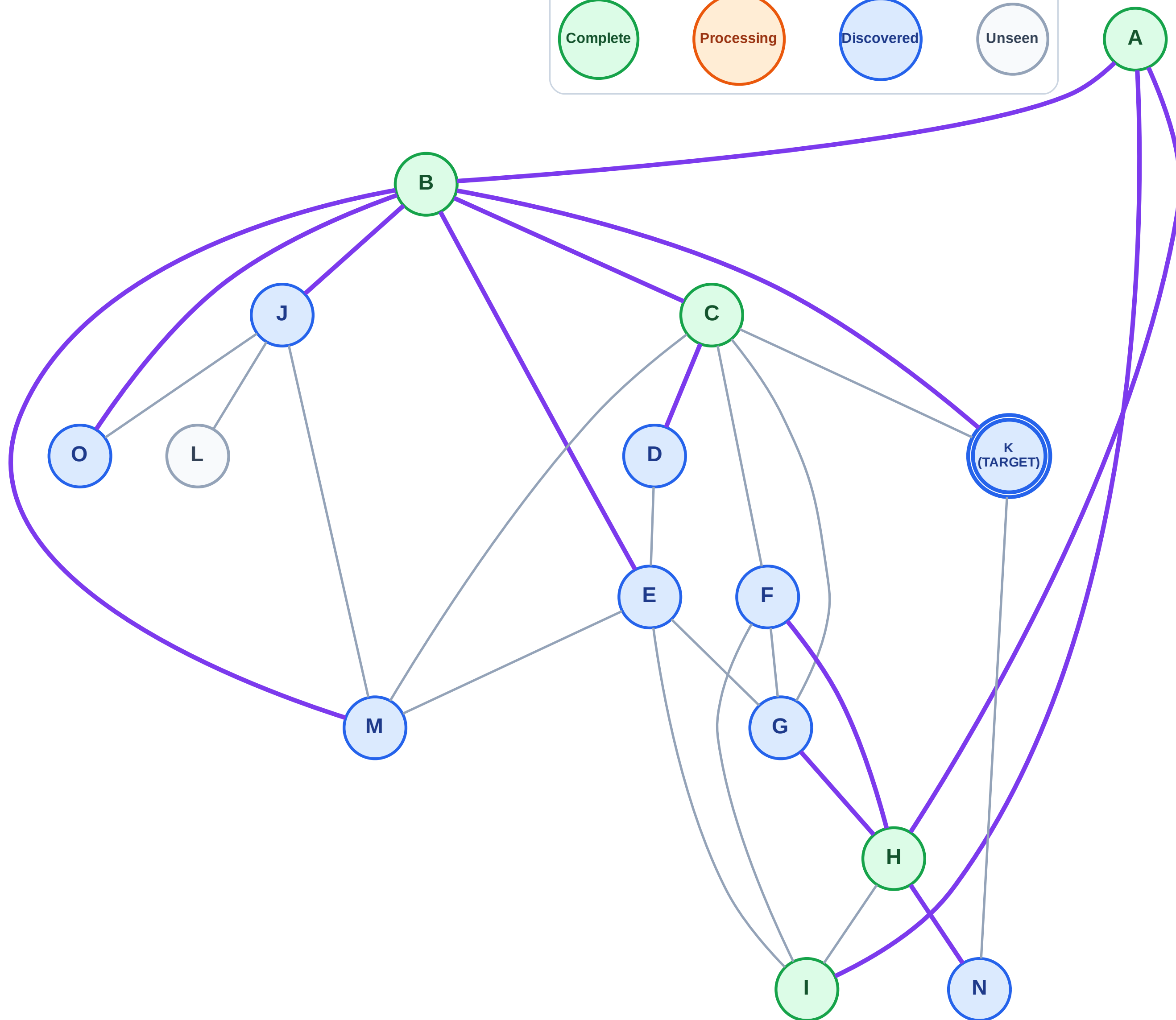
Step 11: Discover D from C

Legend



Queue: E → J → K → M → O → F → G → N → D

Visited: A, B, C, H, I



BFS Traversal

Step 12: Process node E

Legend

Complete

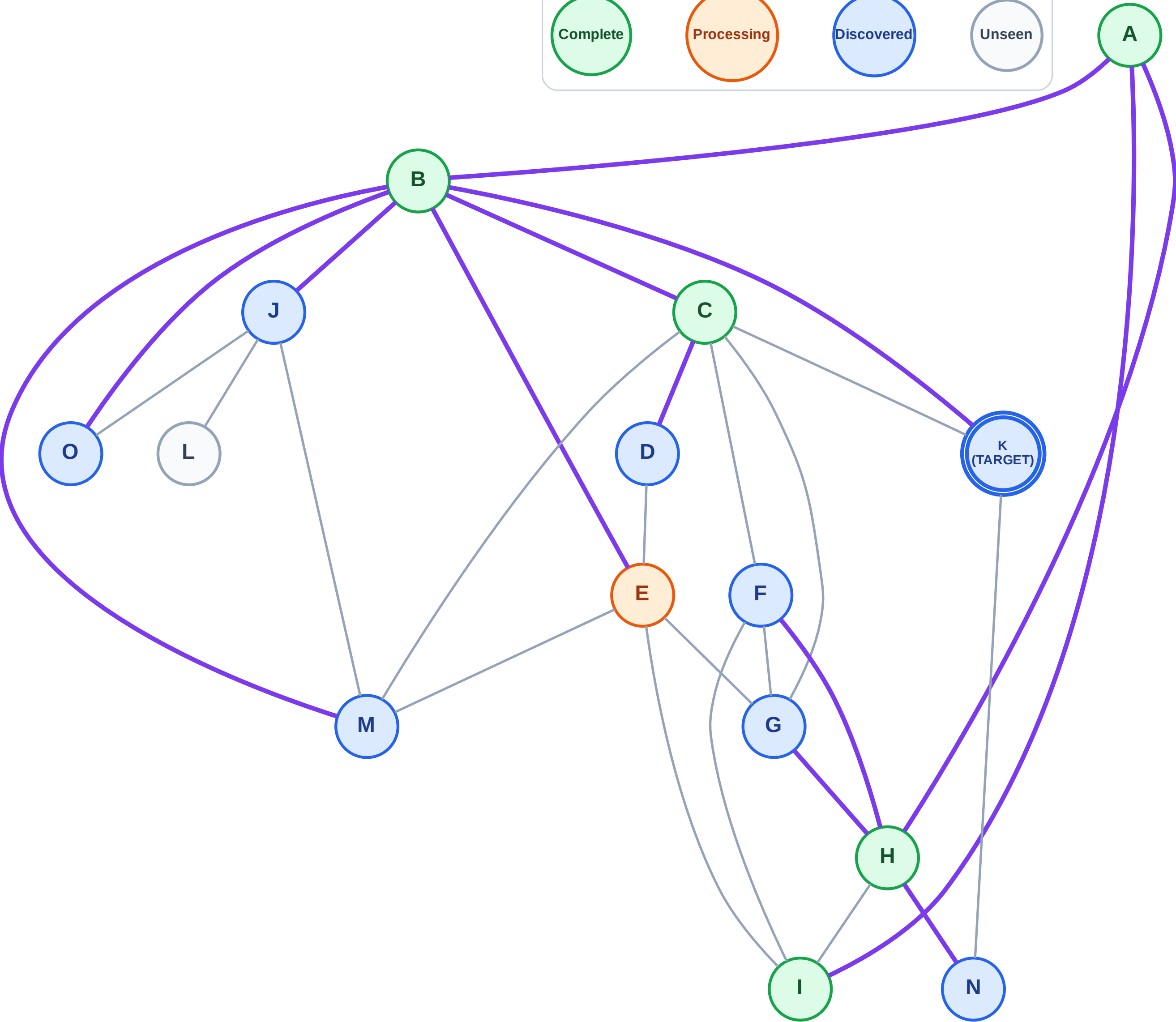
Processing

Discovered

Unseen

Queue: J → K → M → O → F → G → N → D

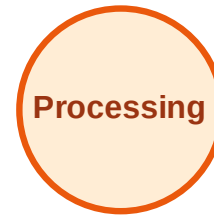
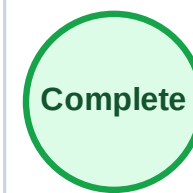
Visited: A, B, C, H, I



BFS Traversal

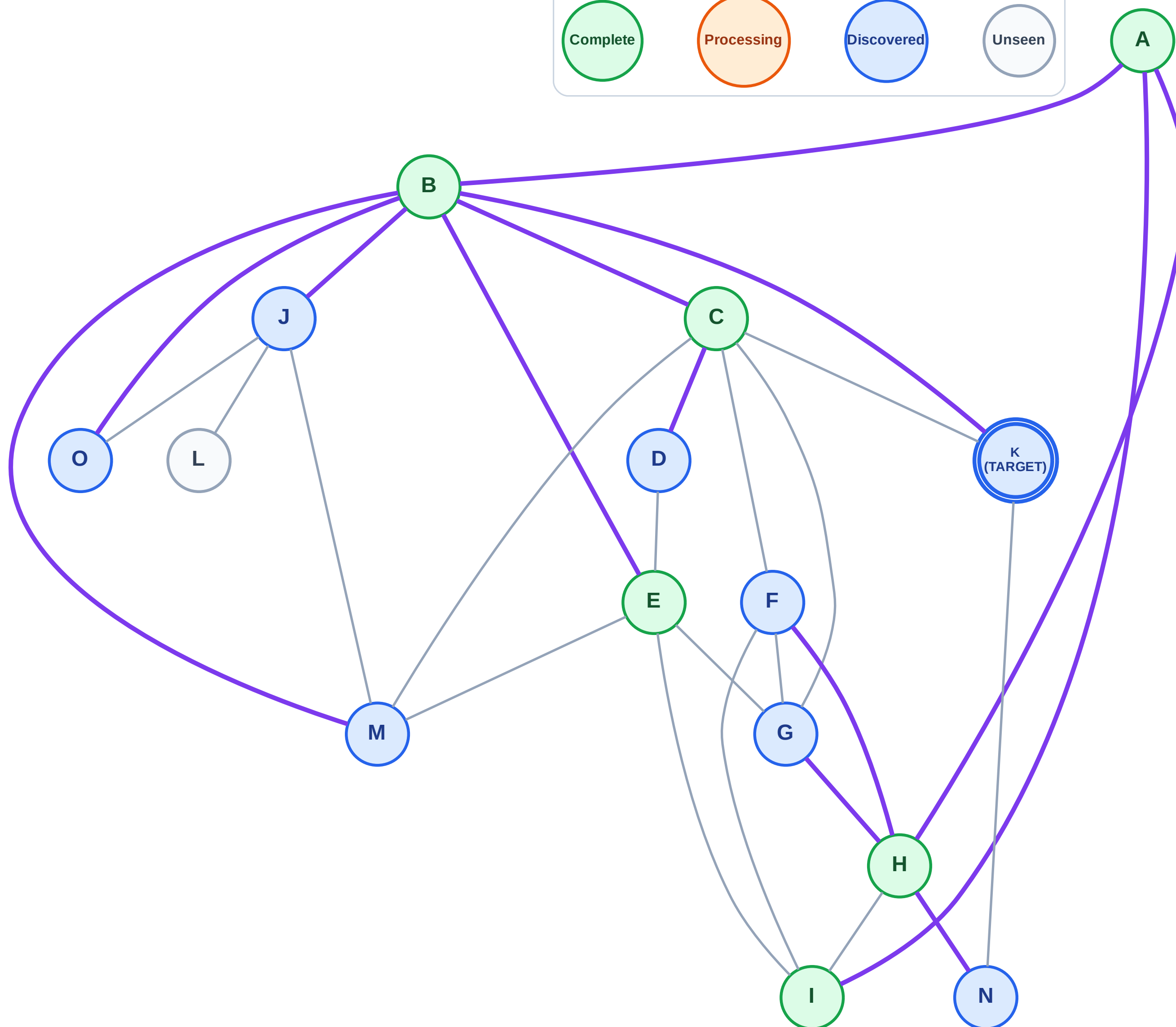
Step 13: No new neighbors from E

Legend



Queue: J → K → M → O → F → G → N → D

Visited: A, B, C, E, H, I



Step 14: Process node J

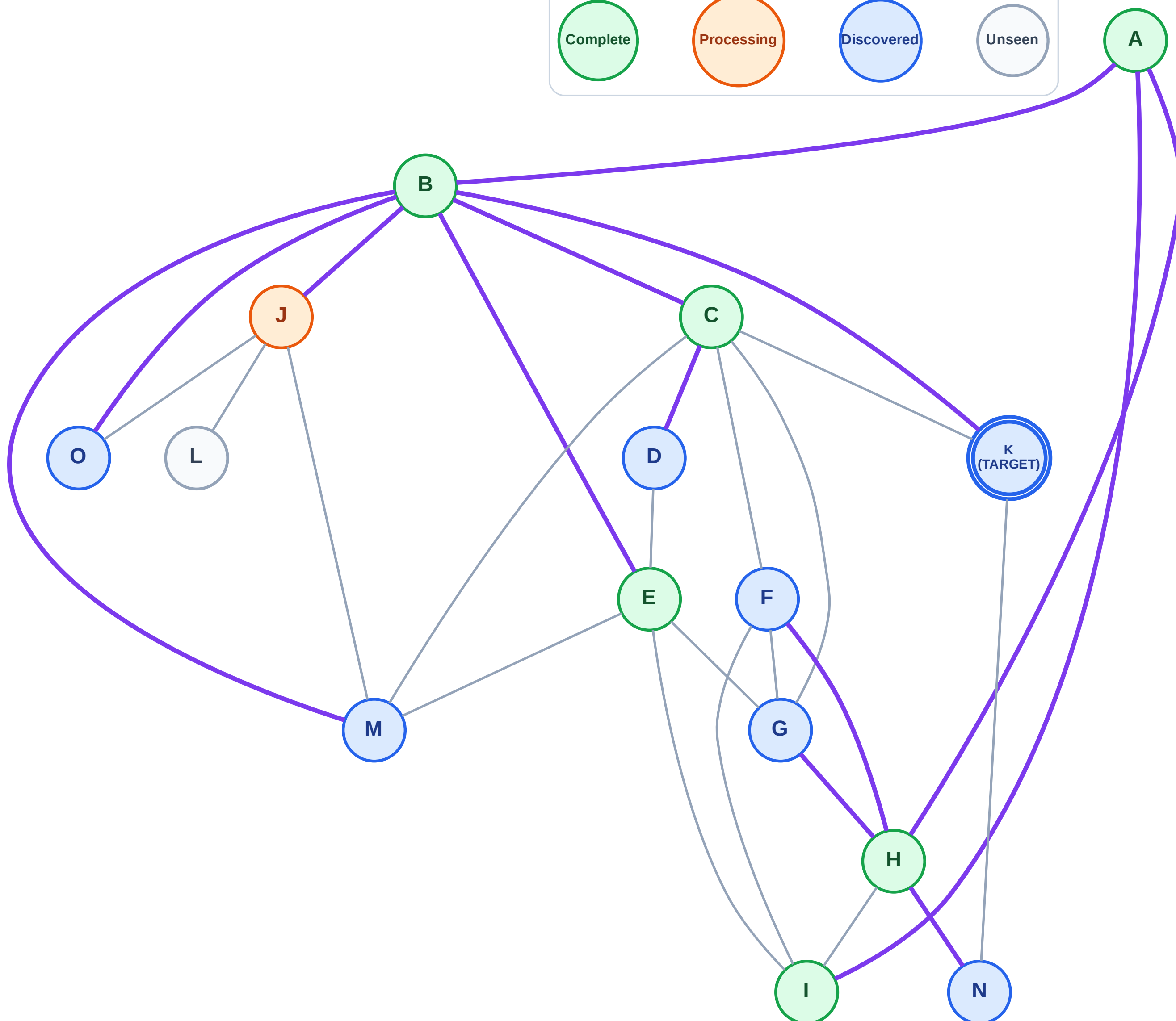
Step 14: Process node J

Legend

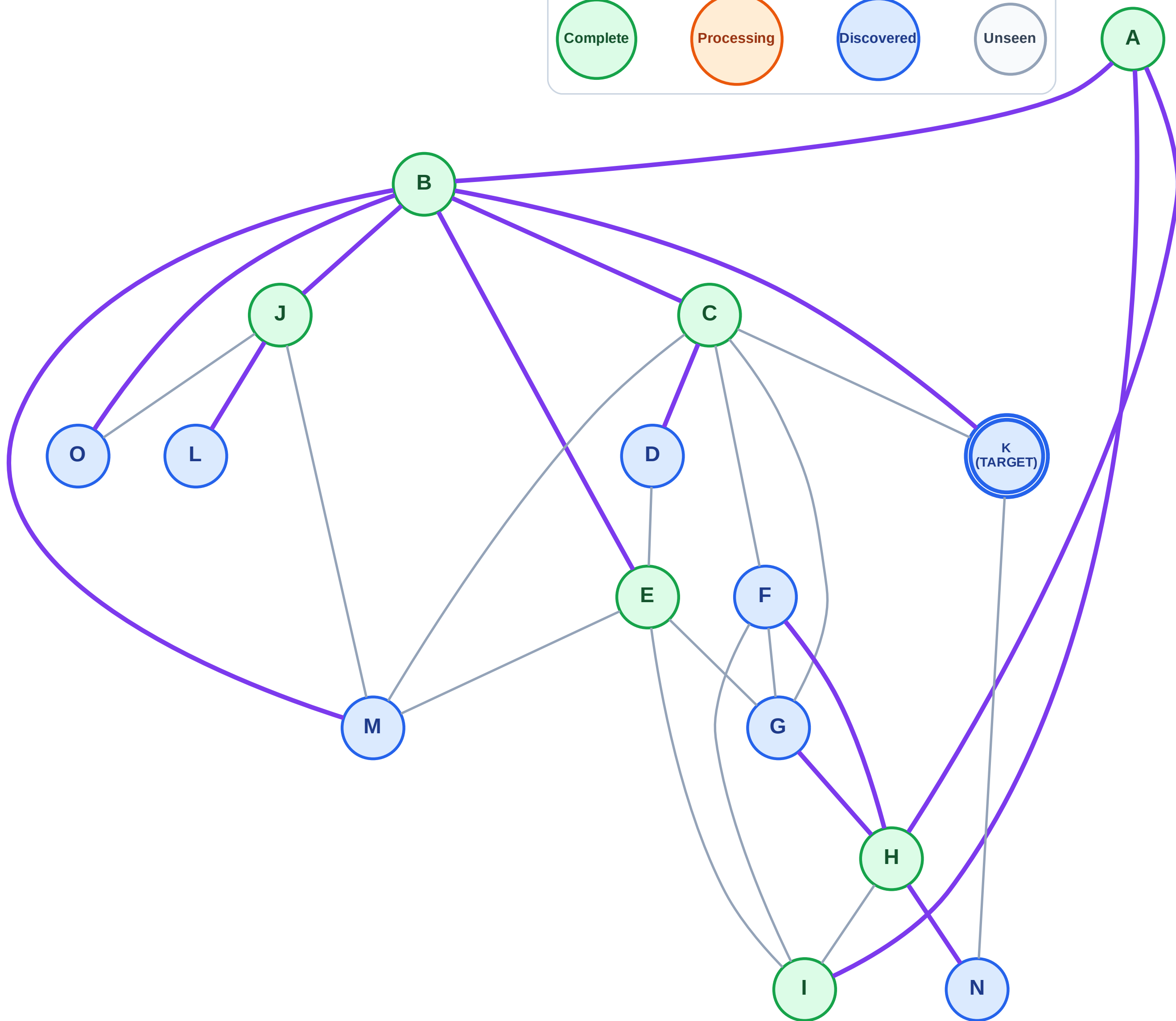
Complete Processing Discovered Unseen

Unseen

Visited: A, B, C, E, H, I



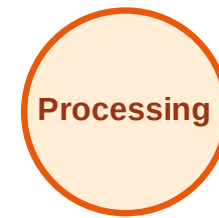
Step 15: Discover L from J



BFS Traversal

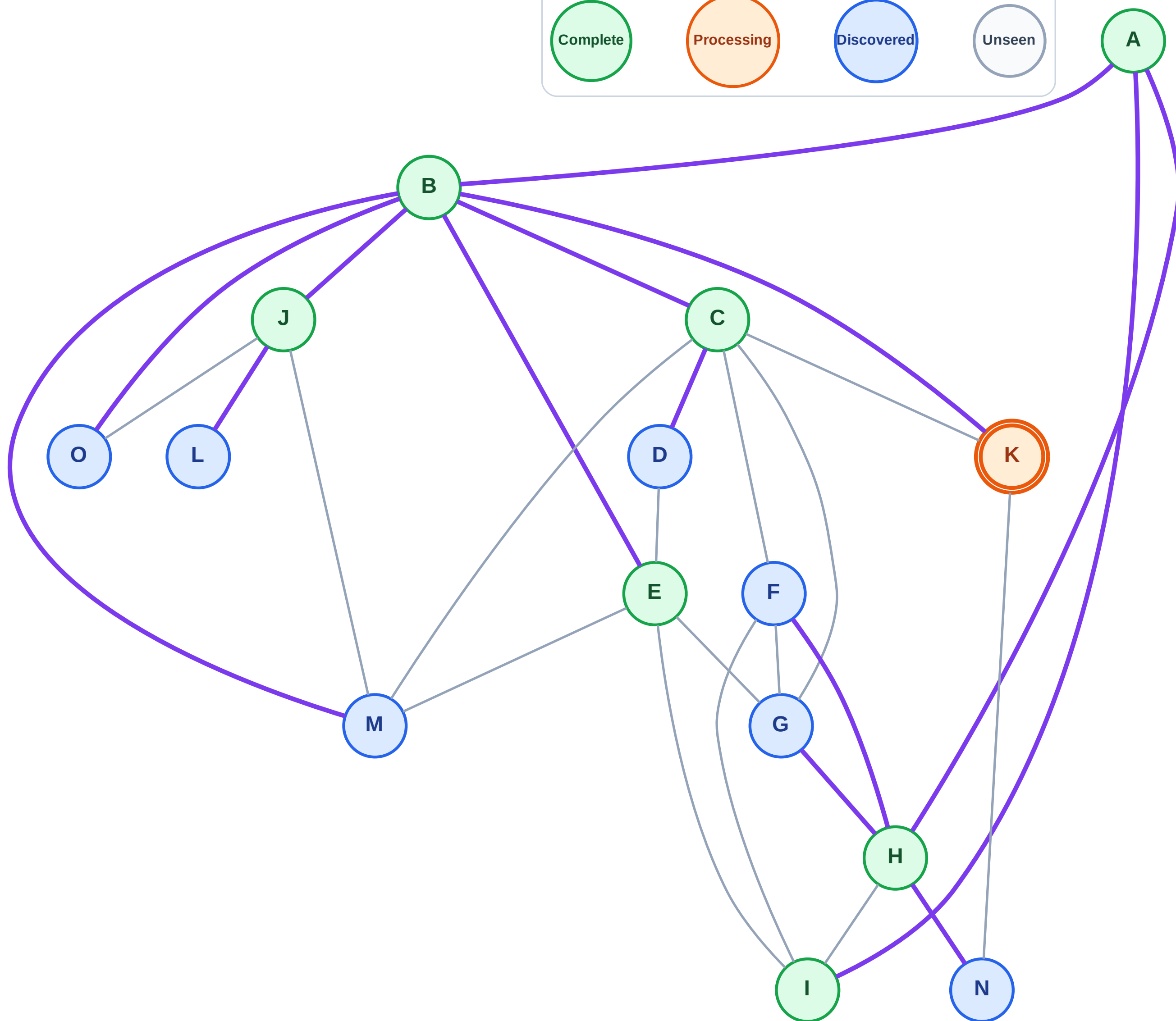
Step 16: Process node K

Legend



Queue: M → O → F → G → N → D → L

Visited: A, B, C, E, H, I, J



BFS Traversal

Step 17: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend

Complete

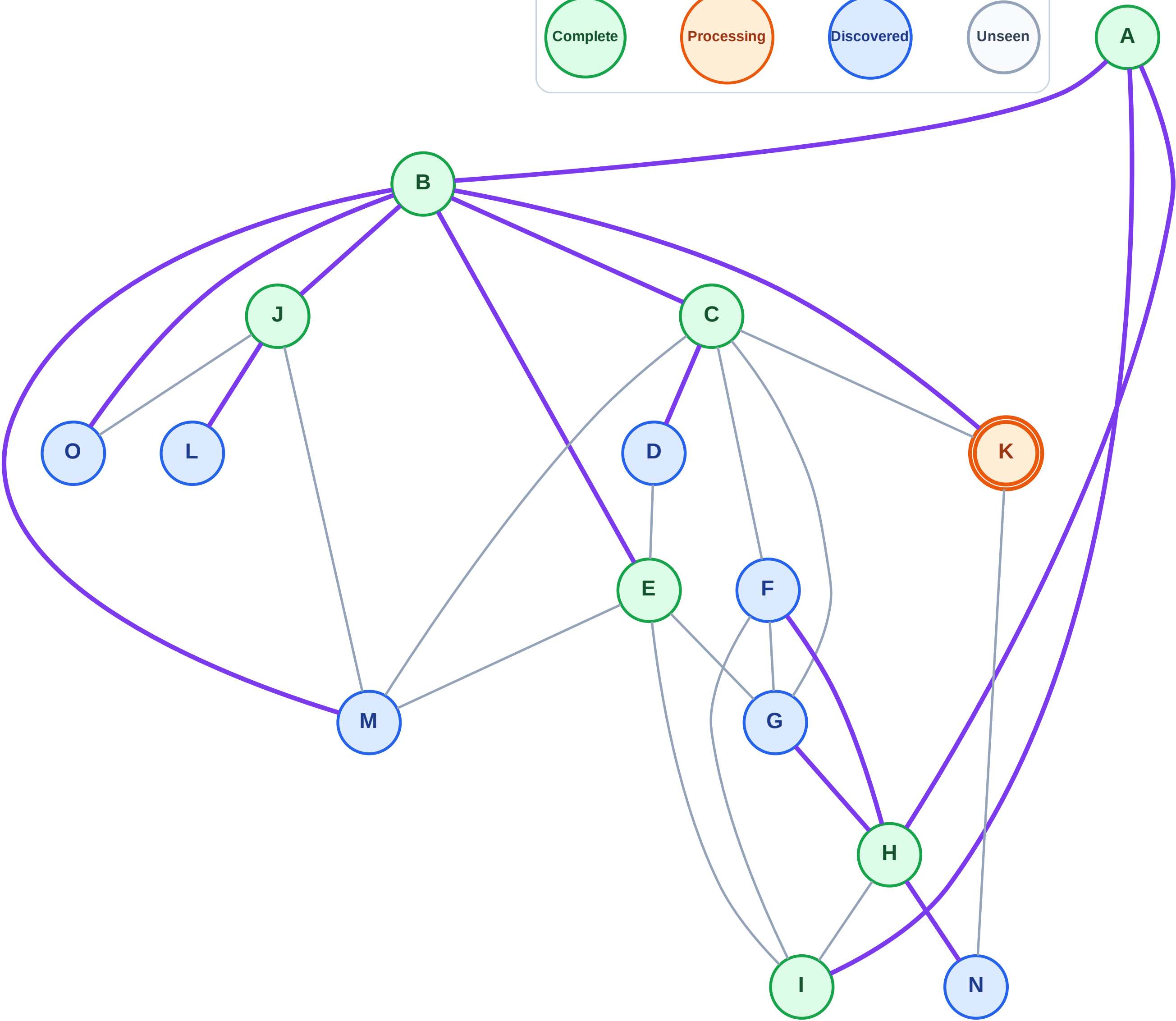
Processing

Discovered

Unseen

Queue: M → O → F → G → N → D → L

Visited: A, B, C, E, H, I, J, K



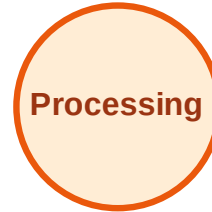
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



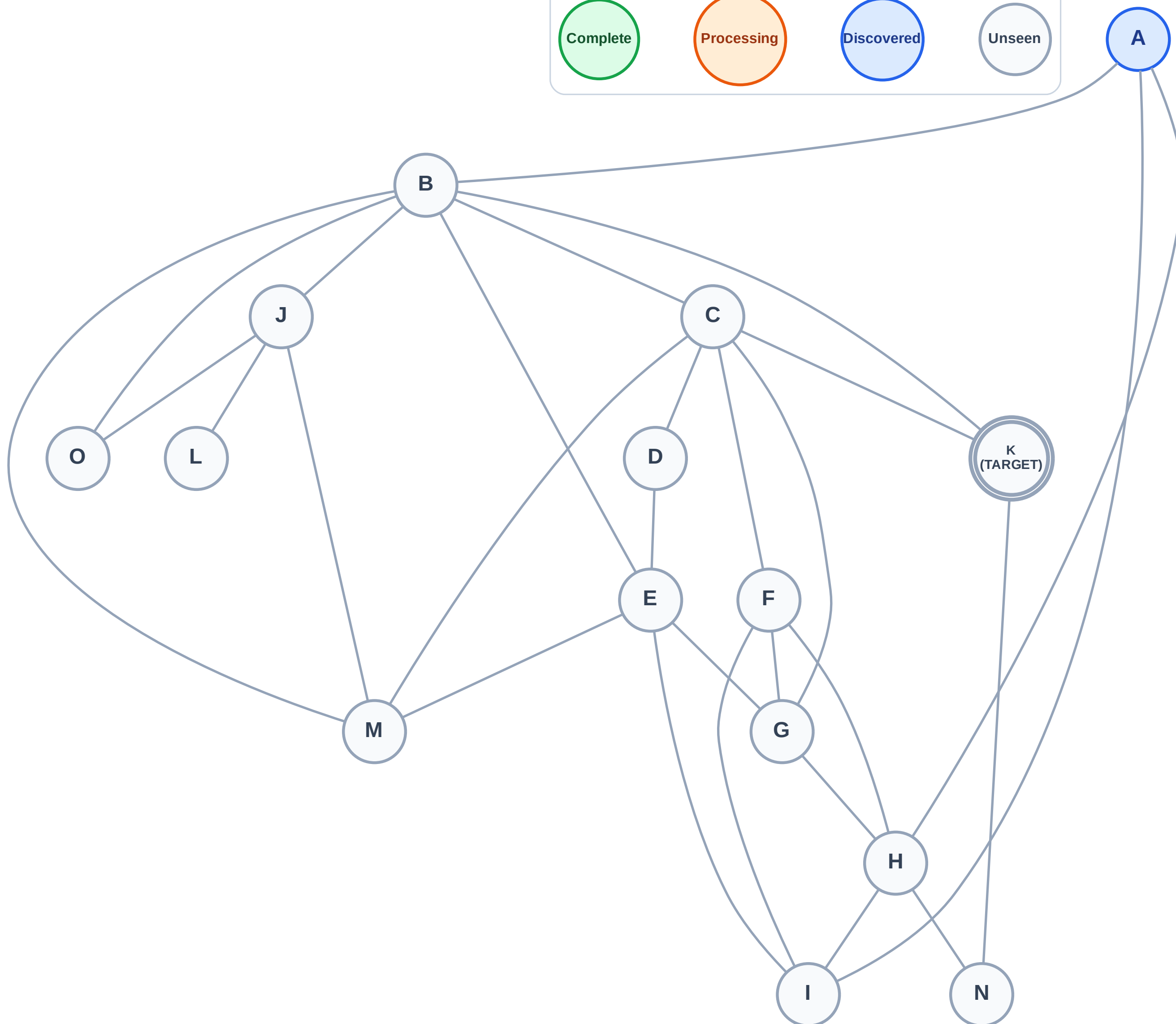
Discovered



Unseen

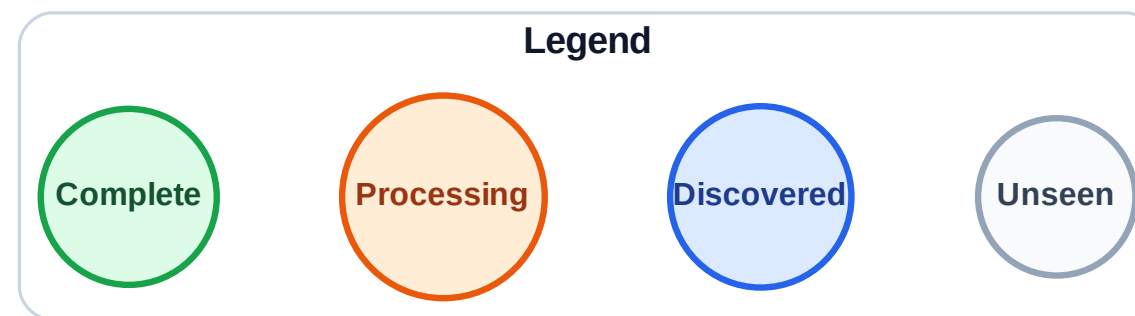
Stack (top at right): A

Visited: (none)



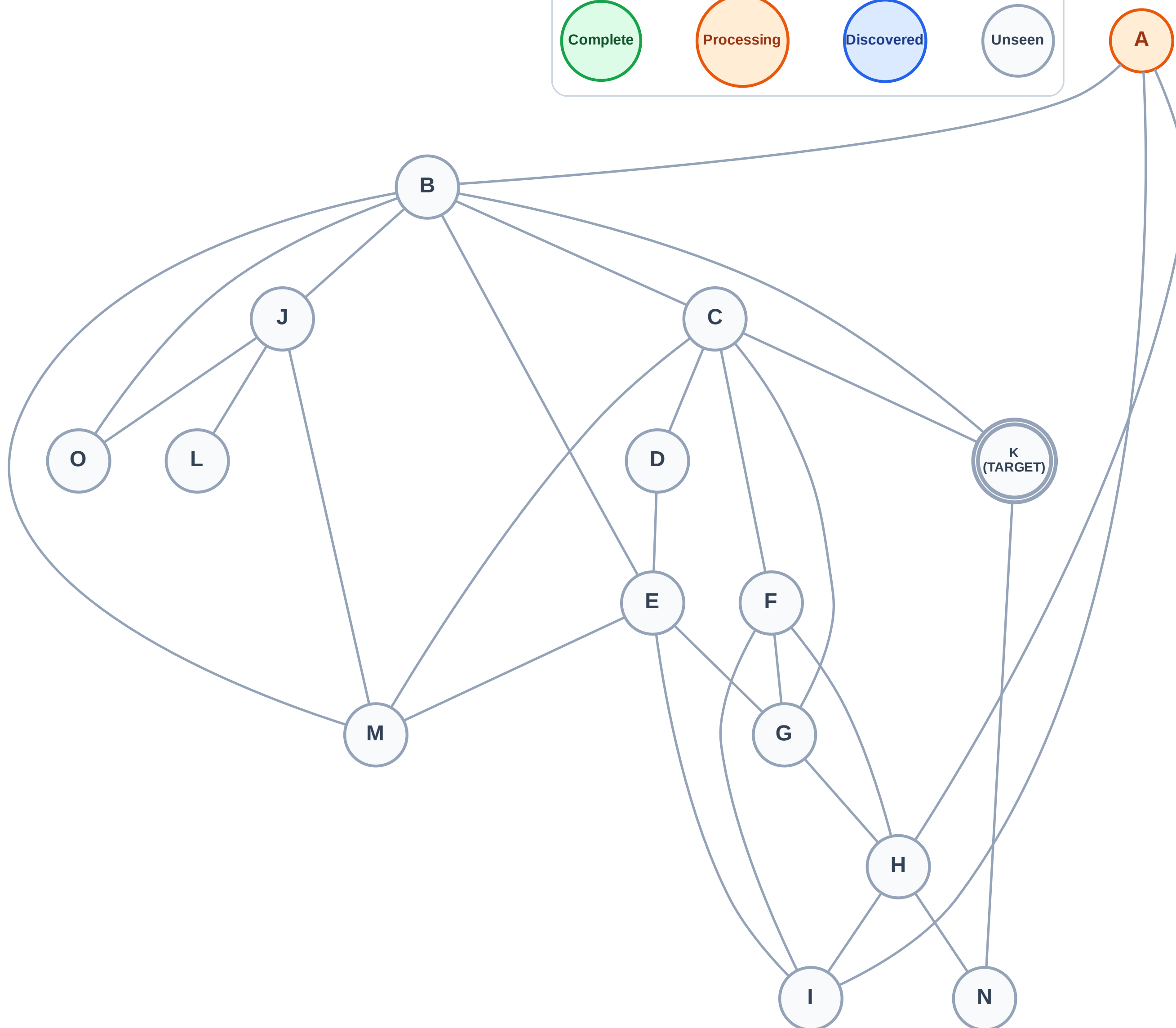
DFS Traversal

Step 2: Process node A



Stack (top at right): (empty)

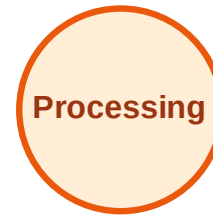
Visited: (none)



DFS Traversal

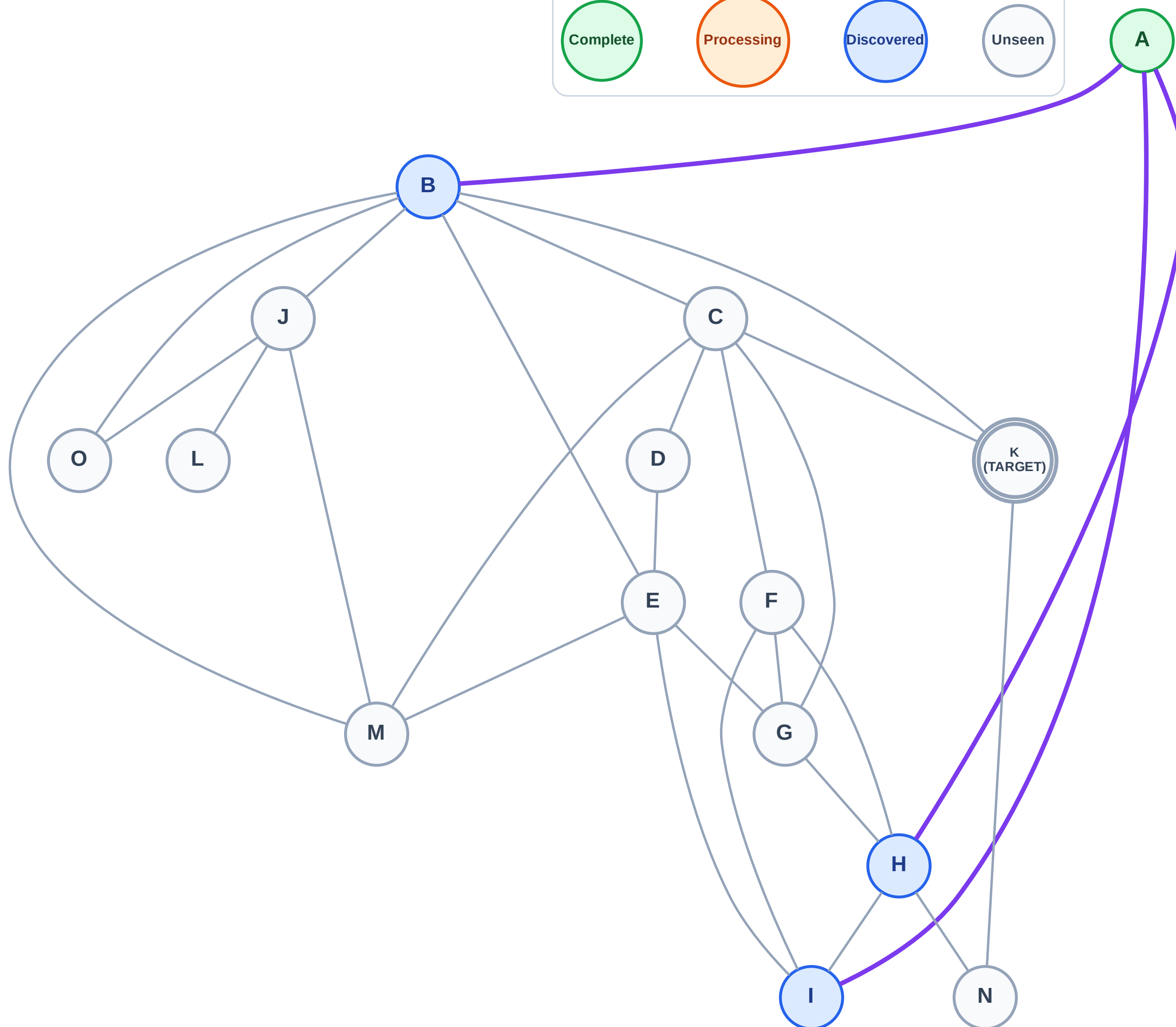
Step 3: Discover I, H, B from A

Legend



Stack (top at right): I | H | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend

Complete

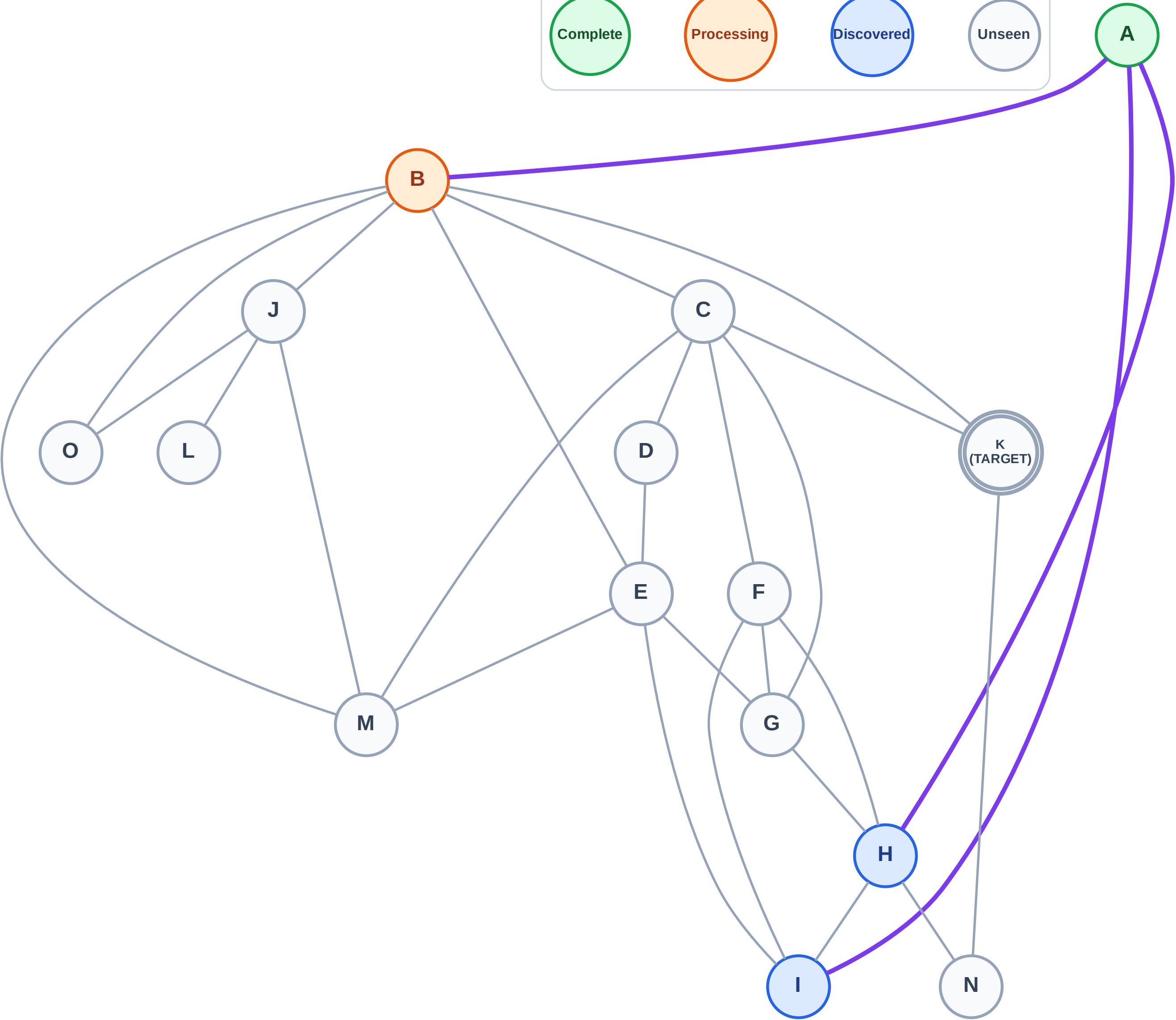
Processing

Discovered

Unseen

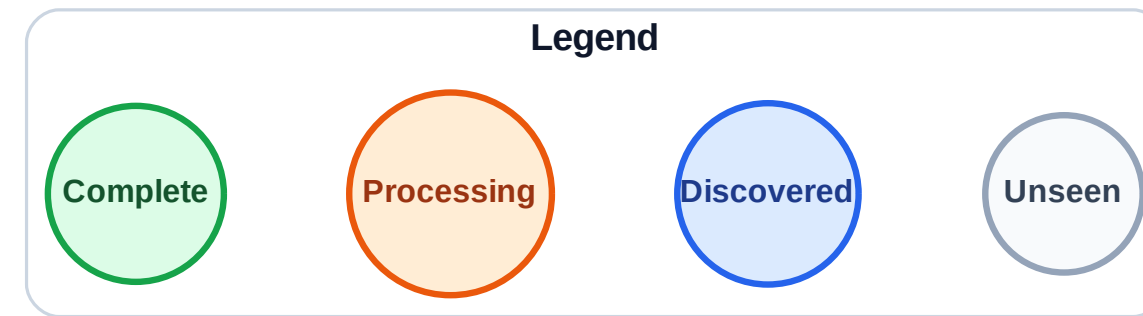
Stack (top at right): I | H

Visited: A



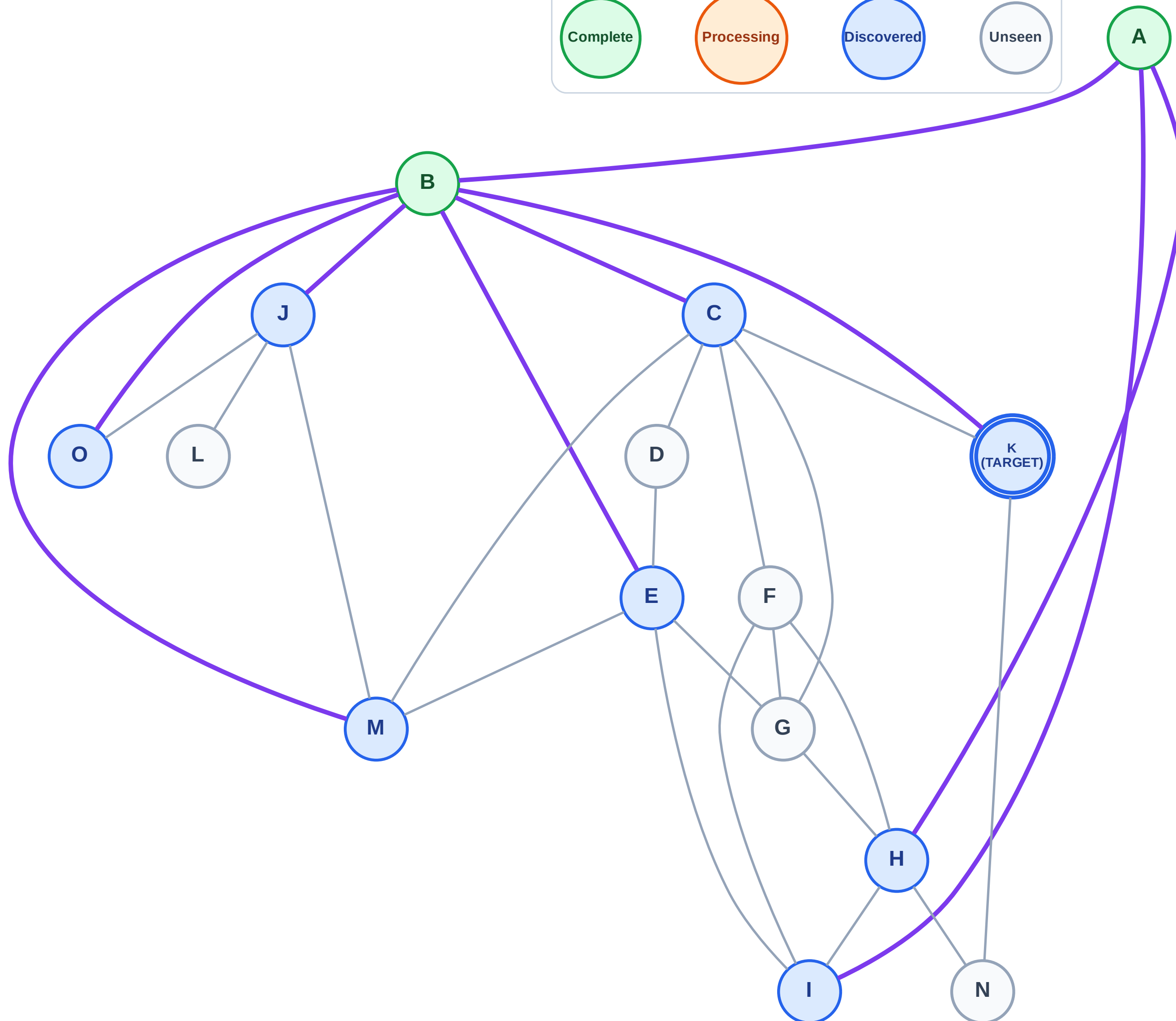
DFS Traversal

Step 5: Discover O, M, K, J, E, C from B



Stack (top at right): I | H | O | M | K | J | E | C

Visited: A, B



DFS Traversal

Step 6: Process node C

Legend

Complete

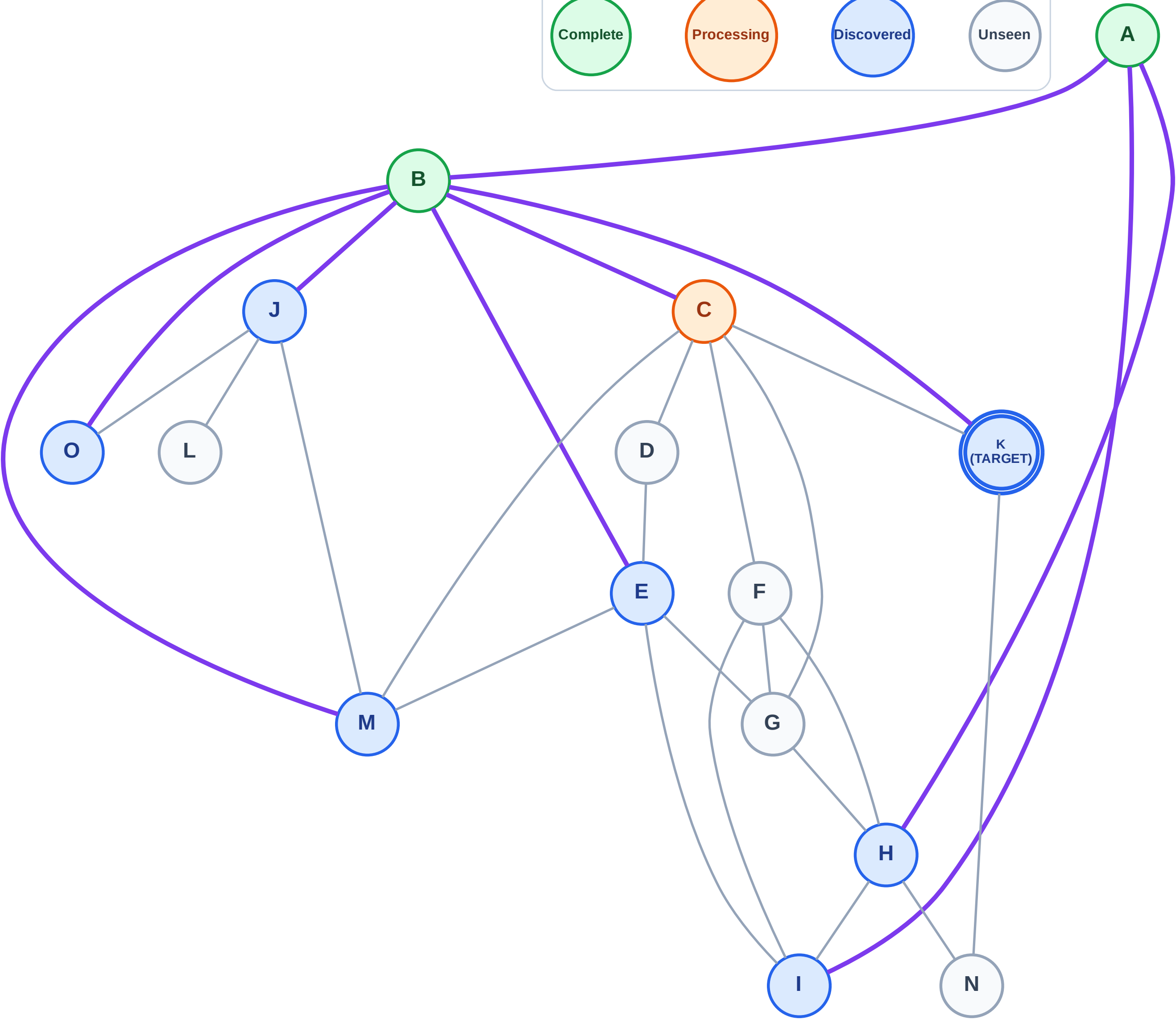
Processing

Discovered

Unseen

Stack (top at right): I | H | O | M | K | J | E

Visited: A, B



Step 7: Discover G, F, D from C

Step 7: Discover G, F, D from C



Visited: A, B, C

Step 8: Process node D

Step 8: Process node D

Legend

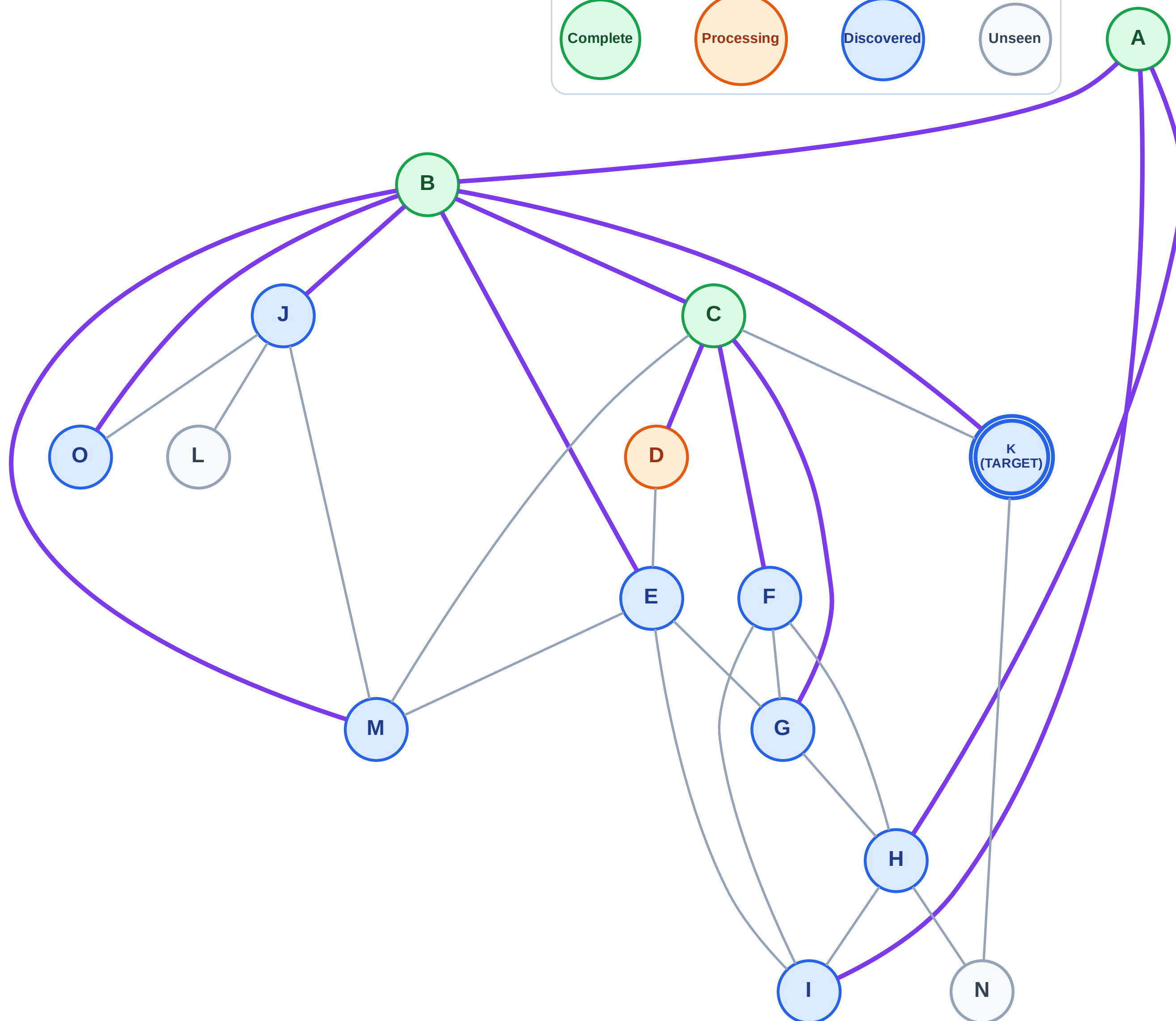
Complete Processing Discovered Unseen

Processing

Unseen

Stack (top at right): I | H | O | M | K | J | E | G | F

Visited: A, B, C



DFS Traversal

Step 9: No new neighbors from D

Legend

Complete

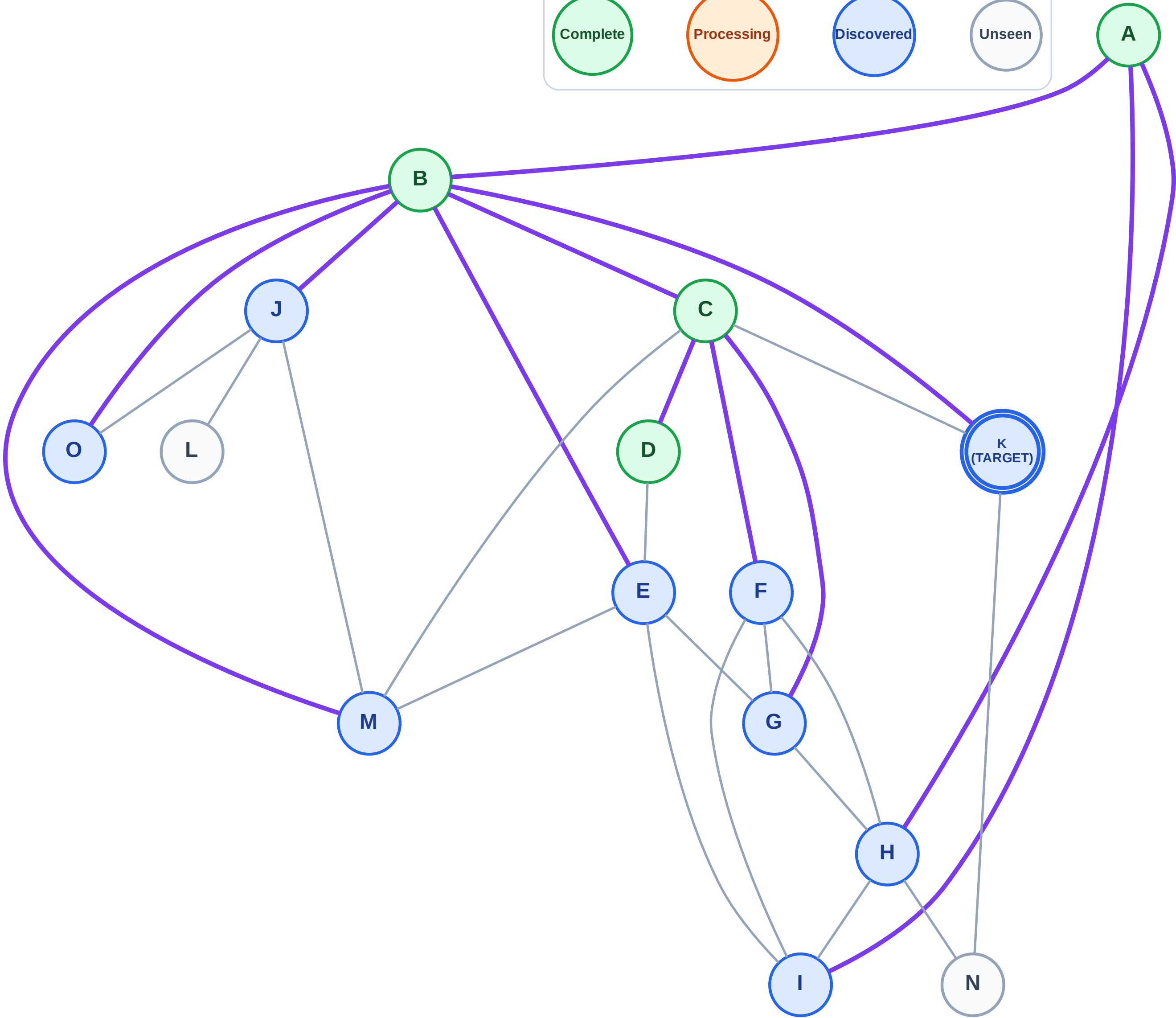
Processing

Discovered

Unseen

Stack (top at right): I | H | O | M | K | J | E | G | F

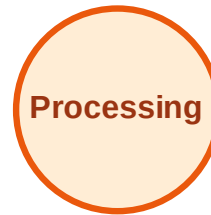
Visited: A, B, C, D



DFS Traversal

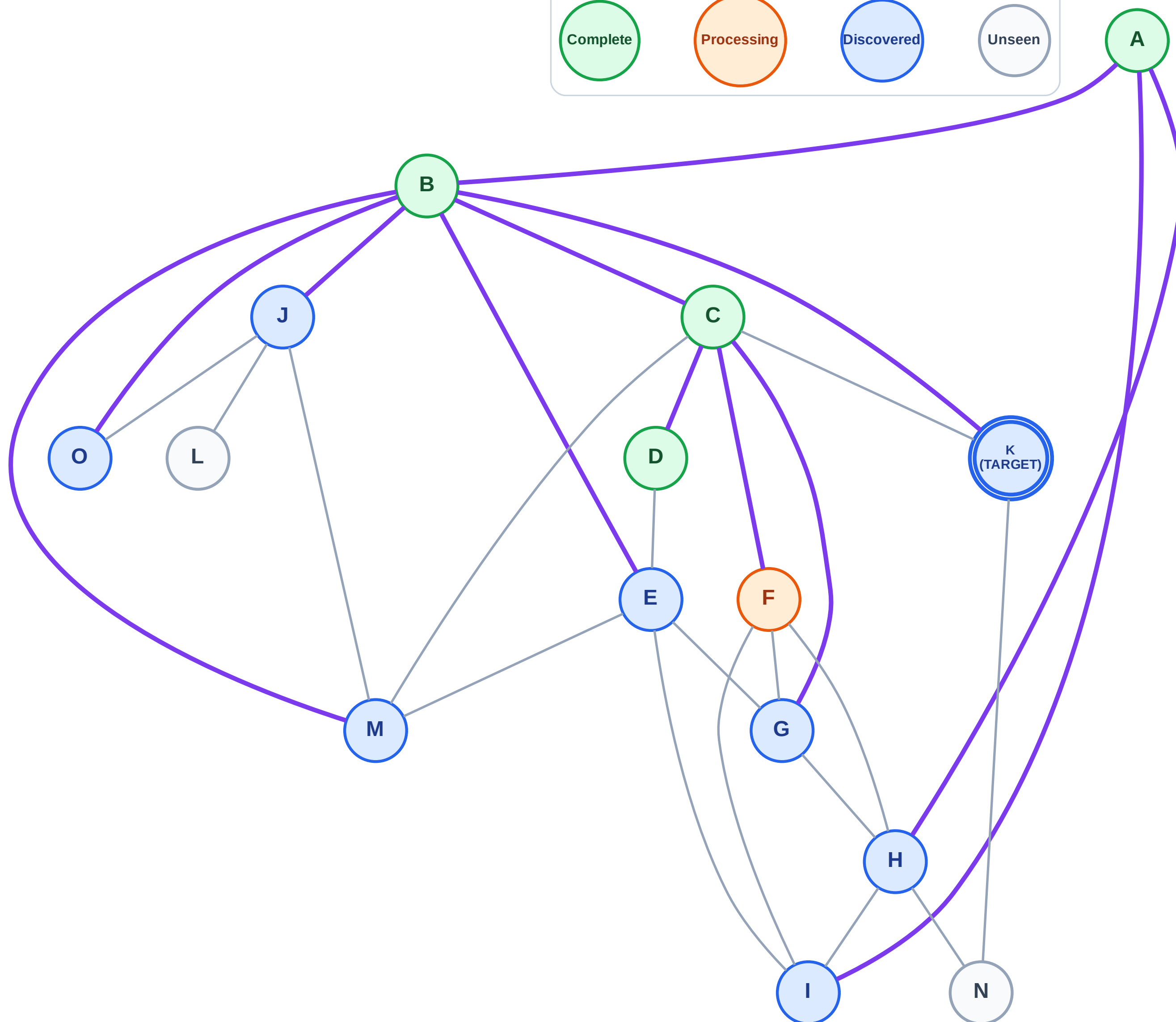
Step 10: Process node F

Legend



Stack (top at right): I | H | O | M | K | J | E | G

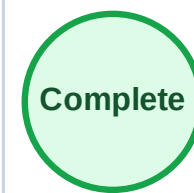
Visited: A, B, C, D



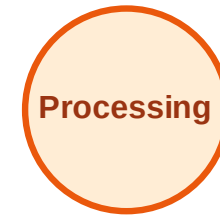
DFS Traversal

Step 11: No new neighbors from F

Legend



Complete



Processing



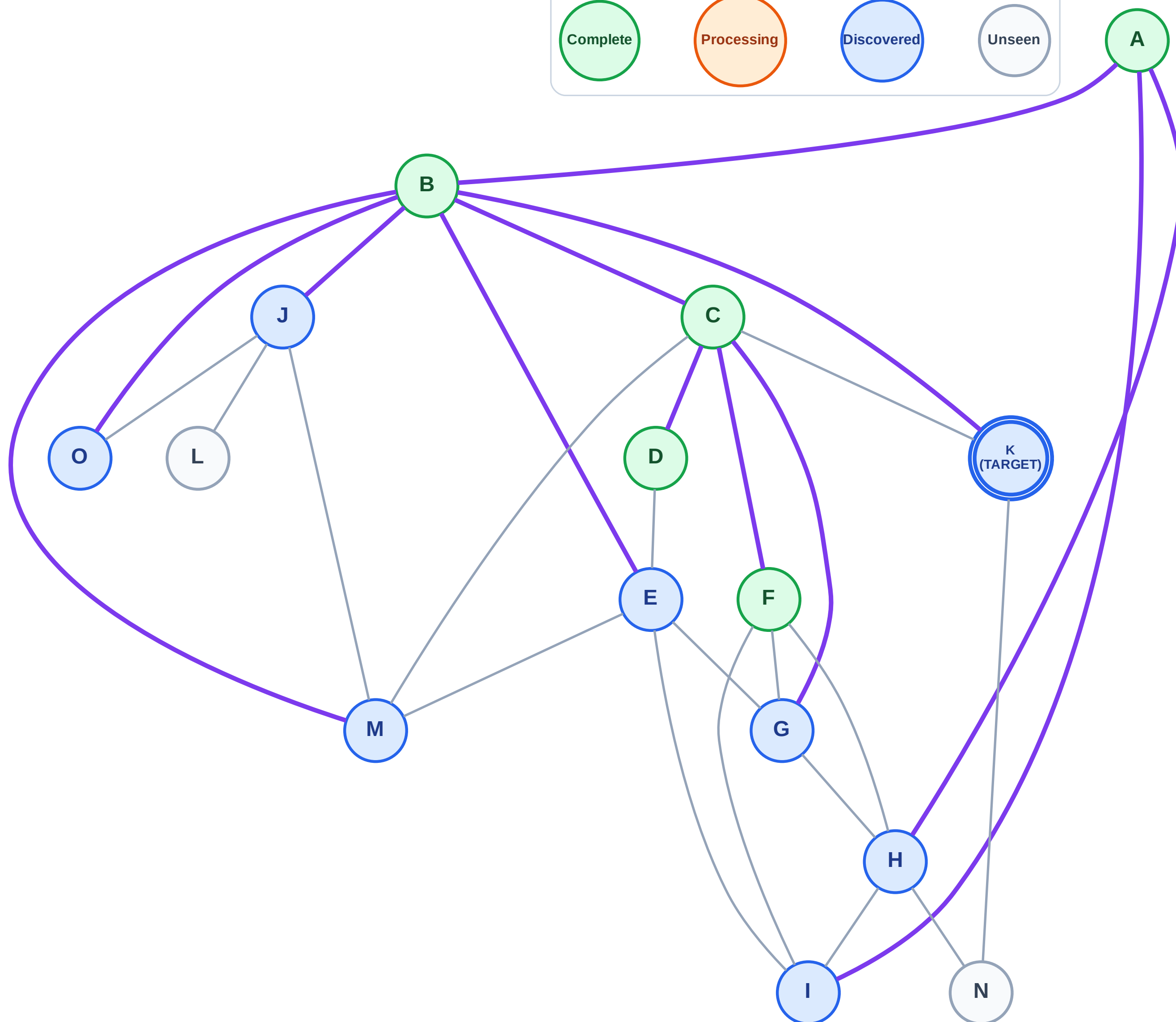
Discovered



Unseen

Stack (top at right): I | H | O | M | K | J | E | G

Visited: A, B, C, D, F



DFS Traversal

Step 12: Process node G

Legend

Complete

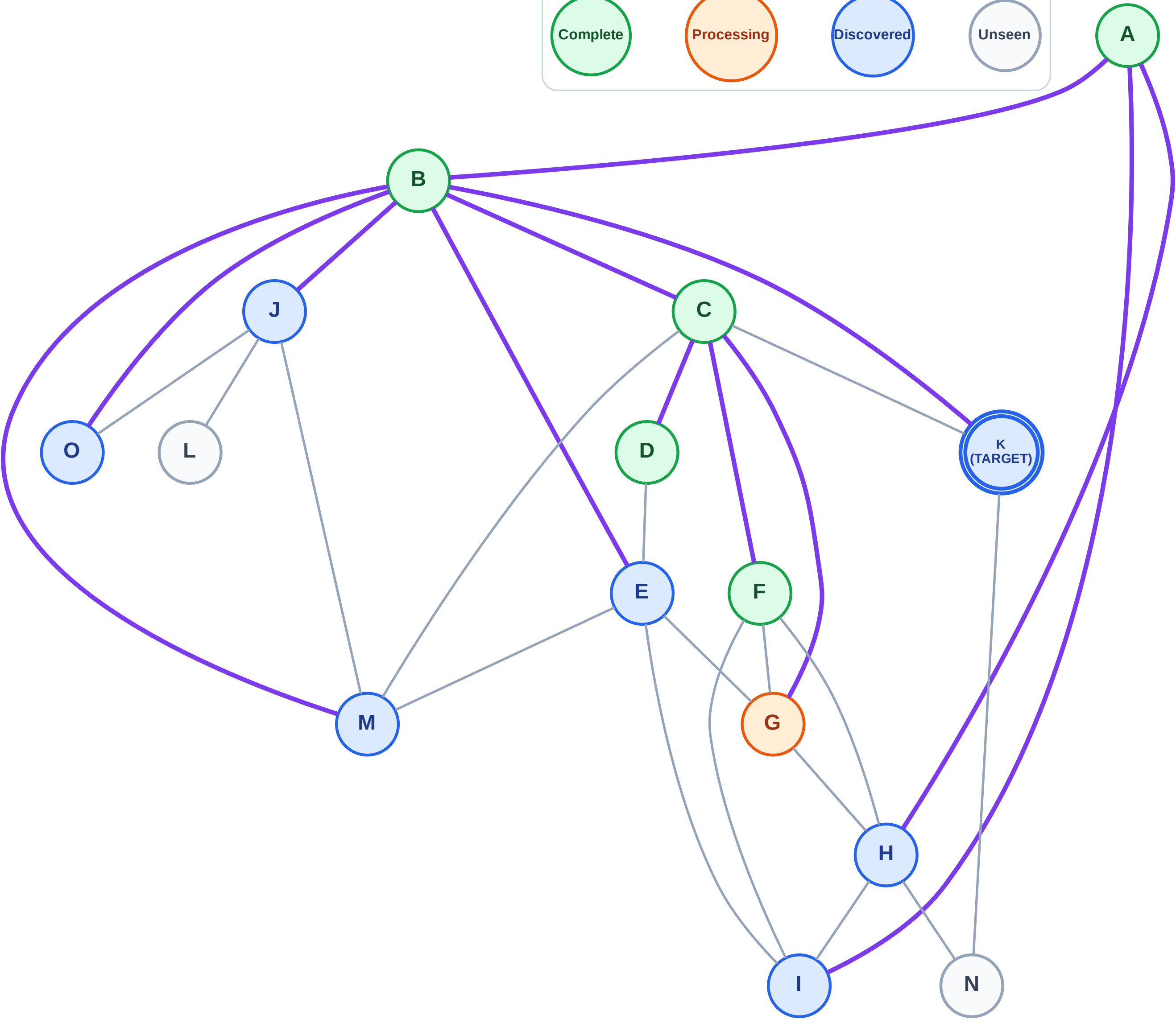
Processing

Discovered

Unseen

Stack (top at right): I | H | O | M | K | J | E

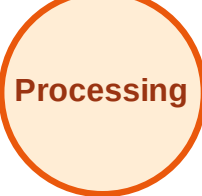
Visited: A, B, C, D, F



DFS Traversal

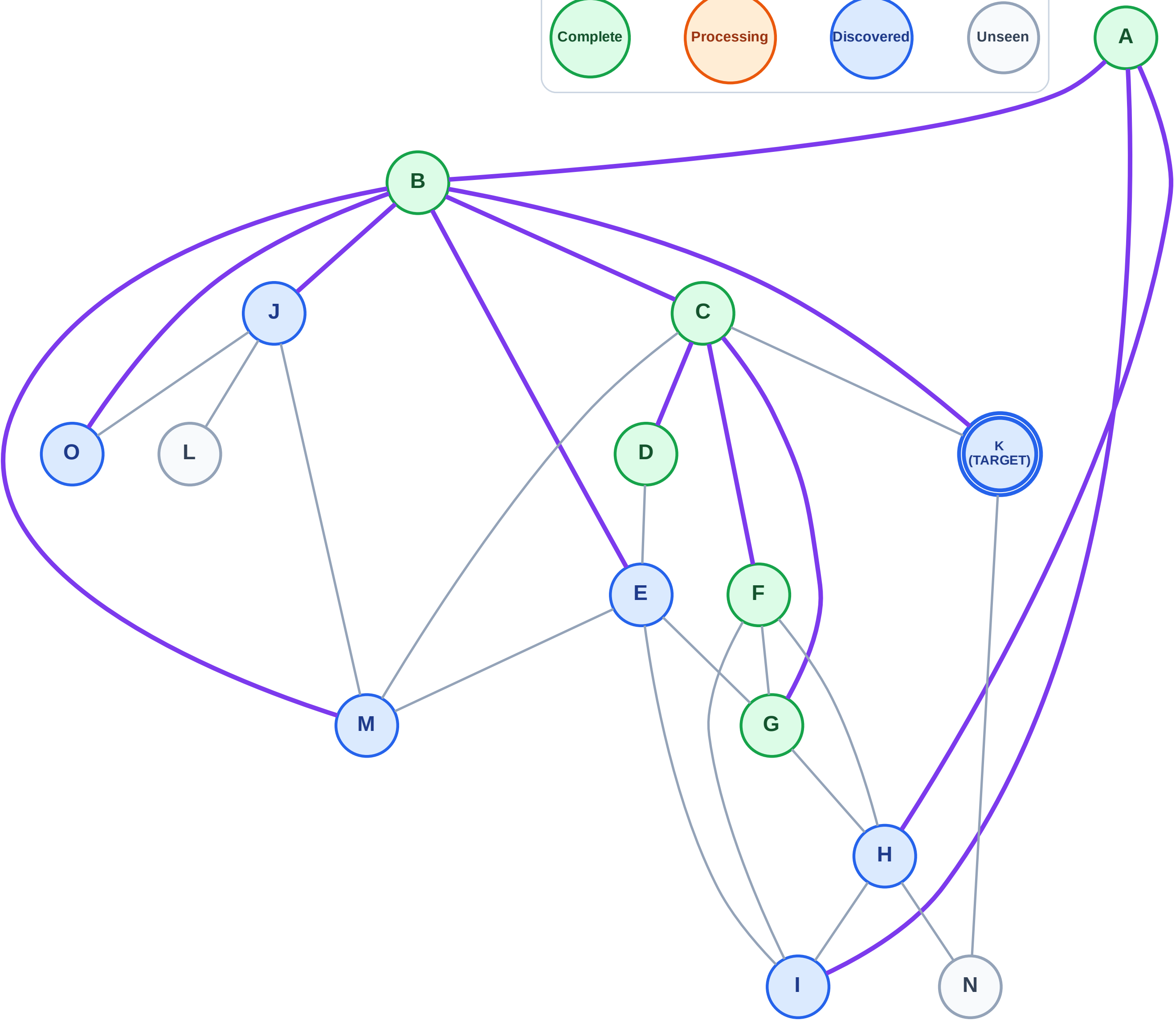
Step 13: No new neighbors from G

Legend



Stack (top at right): I | H | O | M | K | J | E

Visited: A, B, C, D, F, G



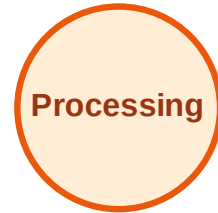
DFS Traversal

Step 14: Process node E

Legend



Complete



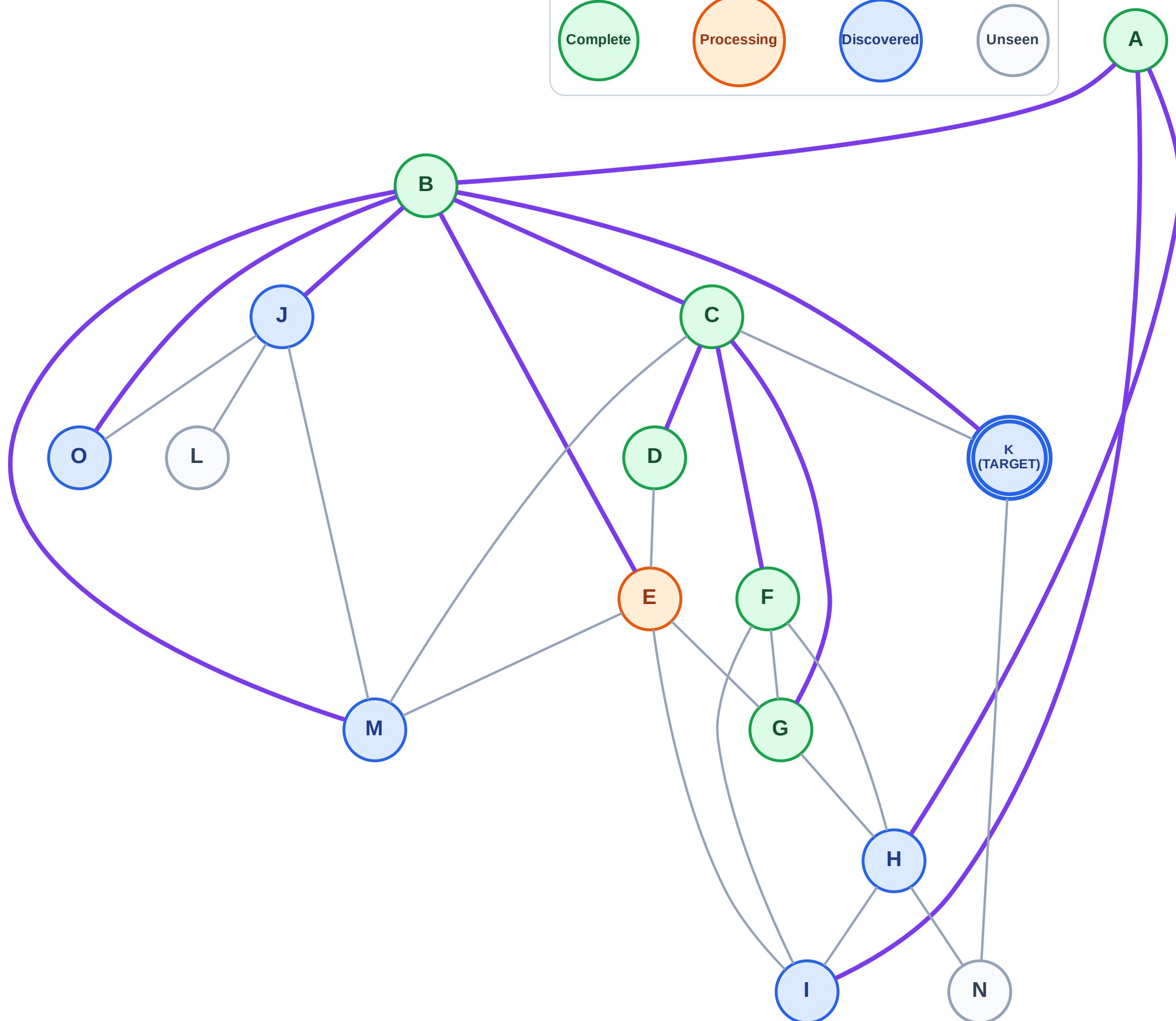
Processing



Discovered



Unseen

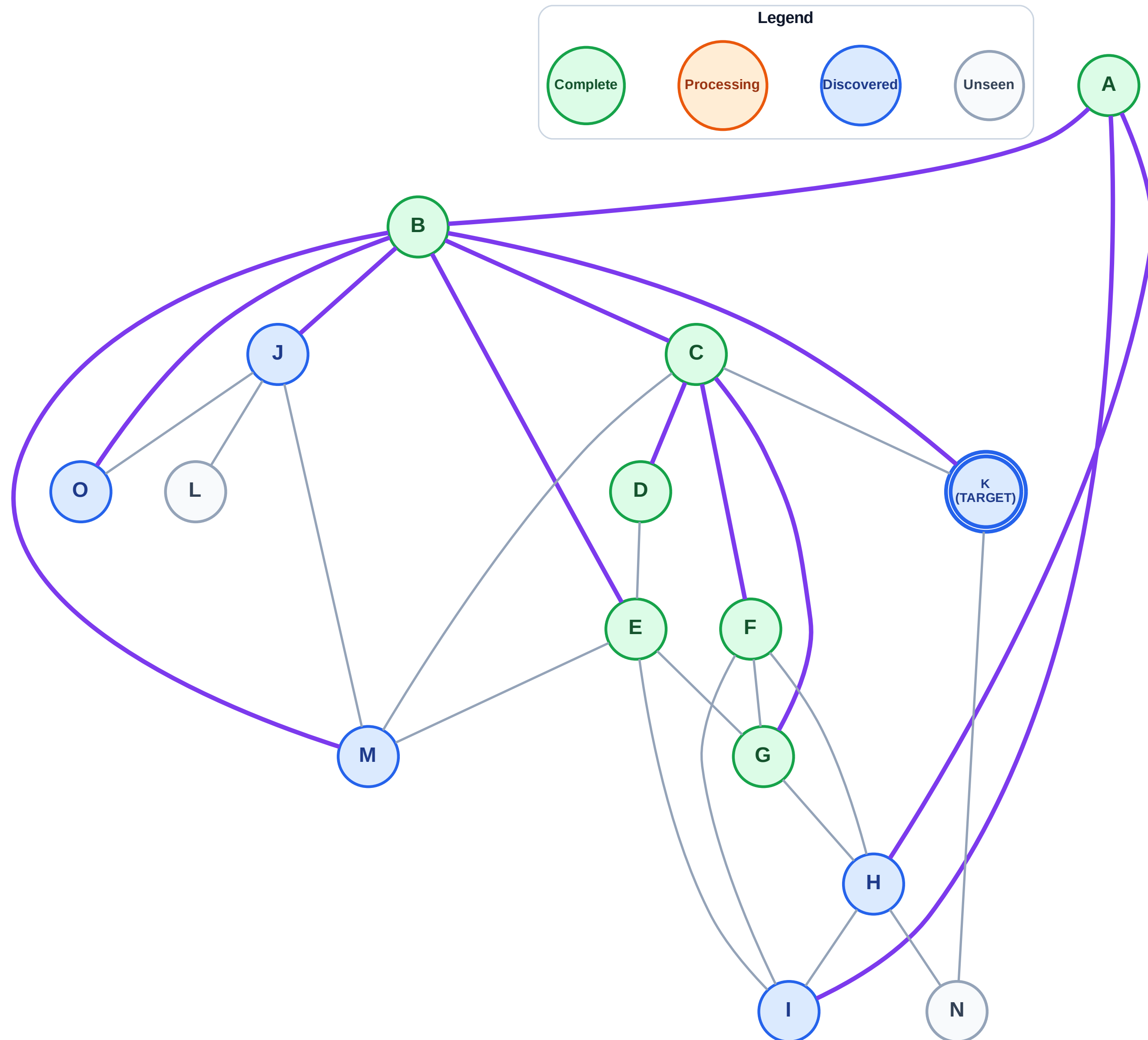


Stack (top at right): I | H | O | M | K | J

Visited: A, B, C, D, F, G

DFS Traversal

Step 15: No new neighbors from E



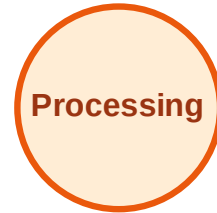
Stack (top at right): I | H | O | M | K | J

Visited: A, B, C, D, E, F, G

DFS Traversal

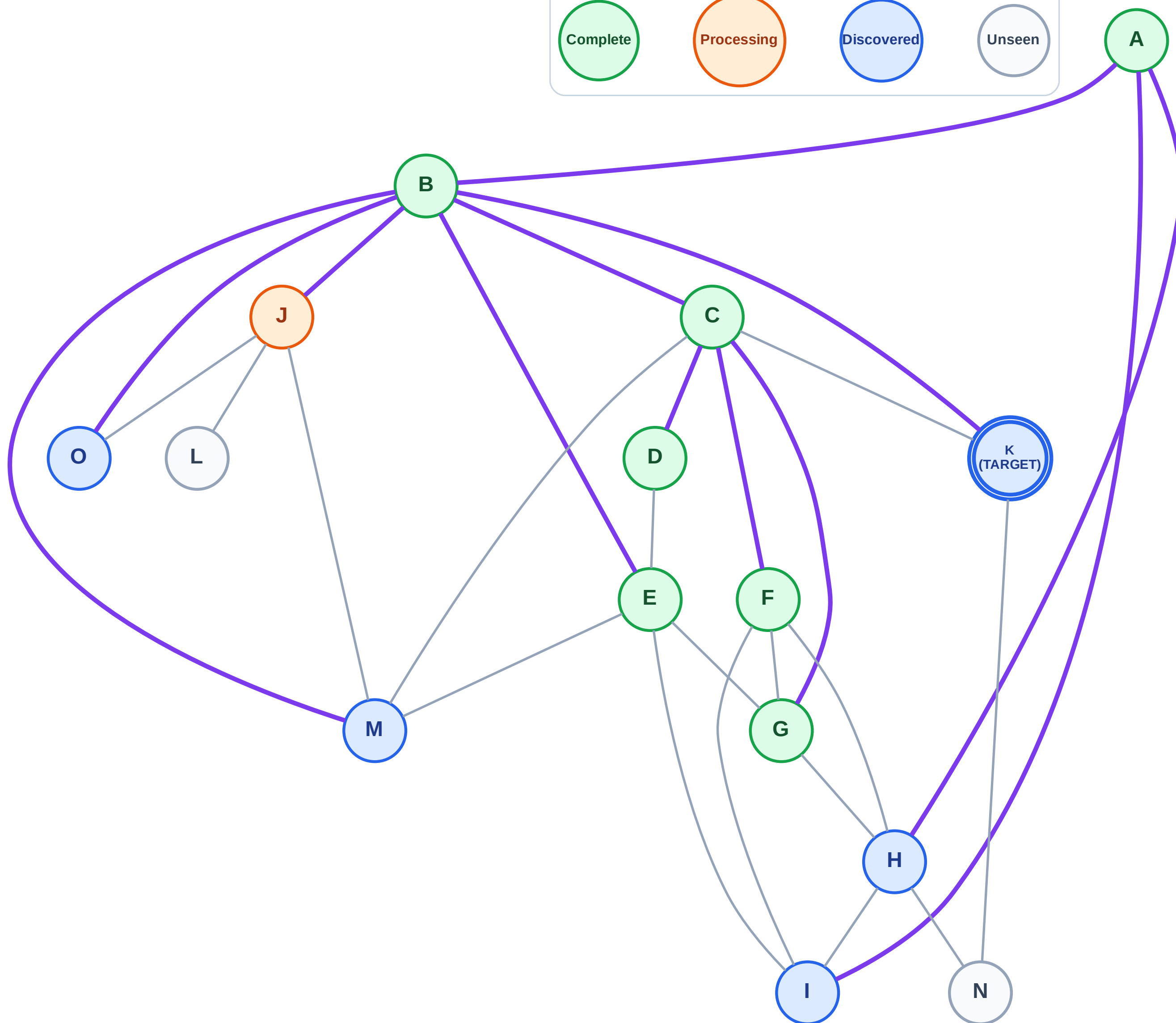
Step 16: Process node J

Legend



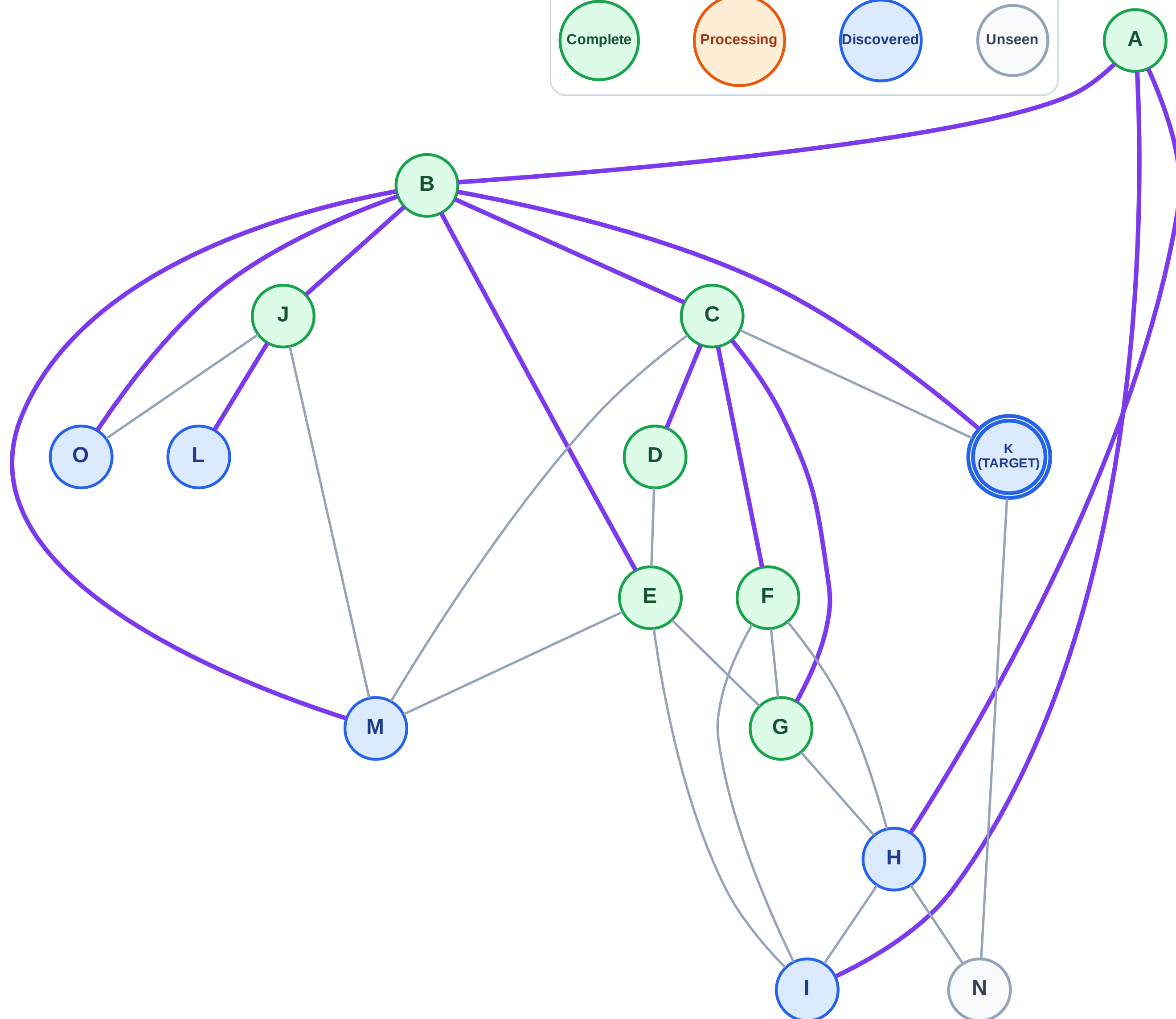
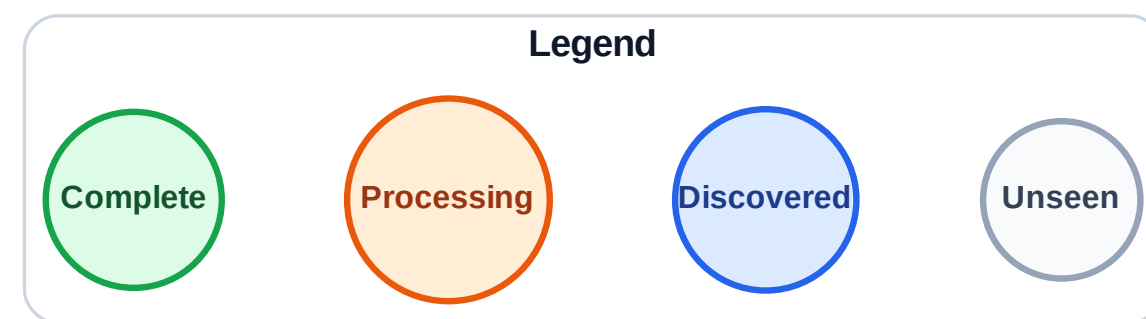
Stack (top at right): I | H | O | M | K

Visited: A, B, C, D, E, F, G



DFS Traversal

Step 17: Discover L from J



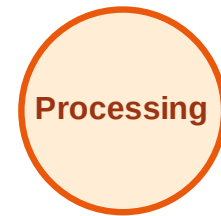
Stack (top at right): I | H | O | M | K | L

Visited: A, B, C, D, E, F, G, J

DFS Traversal

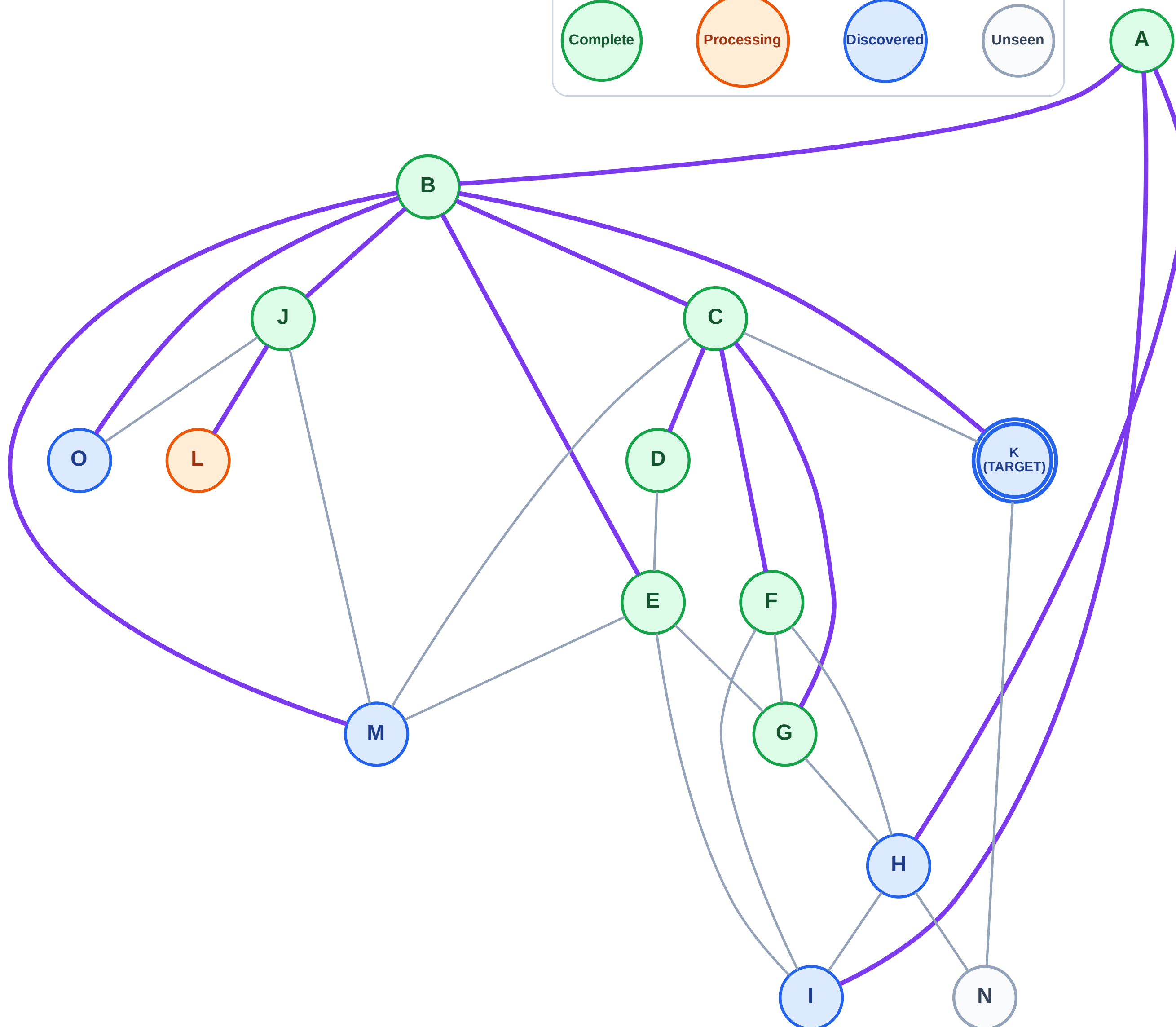
Step 18: Process node L

Legend



Stack (top at right): I | H | O | M | K

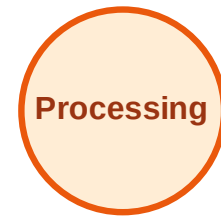
Visited: A, B, C, D, E, F, G, J



DFS Traversal

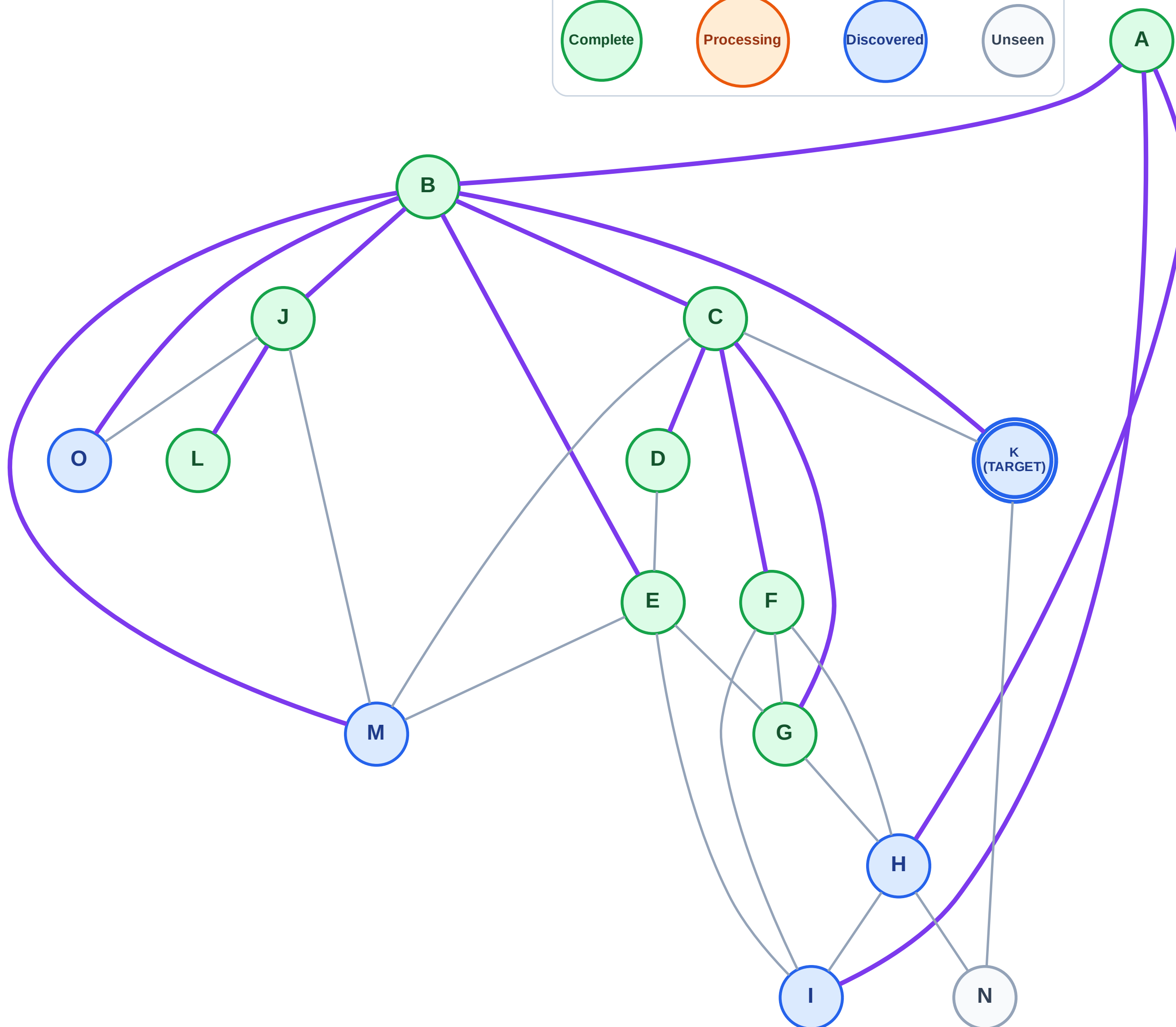
Step 19: No new neighbors from L

Legend



Stack (top at right): I | H | O | M | K

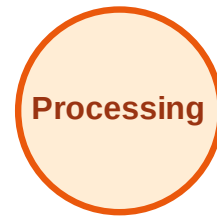
Visited: A, B, C, D, E, F, G, J, L



DFS Traversal

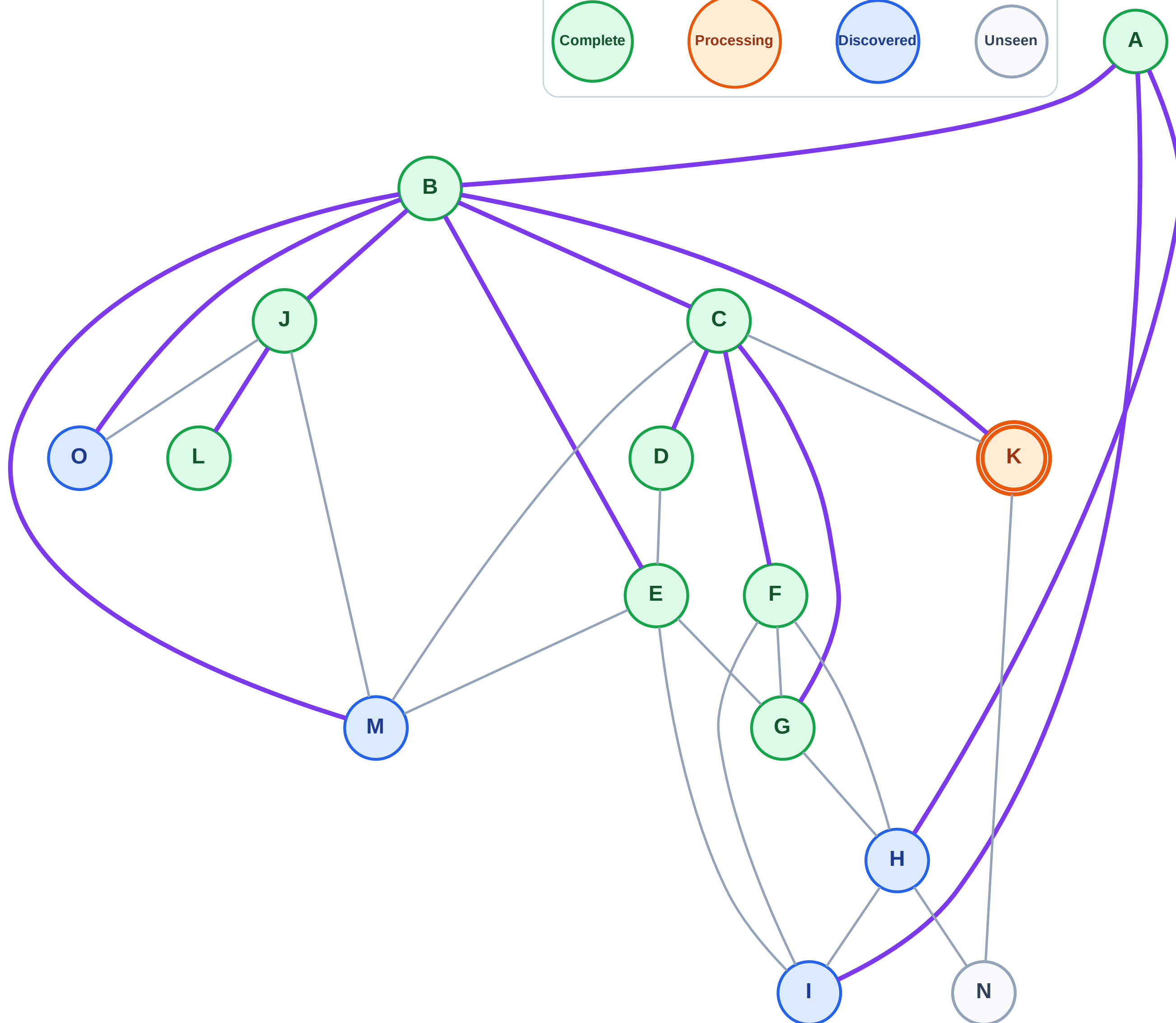
Step 20: Process node K

Legend



Stack (top at right): I | H | O | M

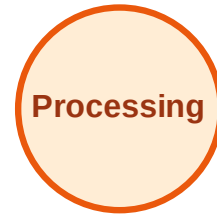
Visited: A, B, C, D, E, F, G, J, L



DFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → B → K

Legend



Stack (top at right): I | H | O | M

Visited: A, B, C, D, E, F, G, J, K, L

